# Priority Design for Grant-Funding Policies

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**Abstract**

Grant-making institutions seek to reward merit while guaranteeing support for designated research categories. We extend priority design from indivisible resources, such as seats or slots, to dividing a budget among grants that differ in cost and may be partially funded. Categories can be supported by earmarking funds or by setting floors. With earmarks, we characterize the unique rule that is non-wasteful, respects merit priorities, and splits each grant's cost equally among its categories wherever their reserves allow. With floors, which minimum costs can make unattainable, we propose a relaxation and a simple rule that satisfies it. Under this rule, requesting more funding at the same merit cannot turn a rejection into an award, and grants never in contention do not affect the allocation. By contrast, on integer floor problems, the rule that funds the highest-merit grants as generously as the constraints allow loses both properties.

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# 1 Introduction

Support for scientific research in the United States is largely apportioned via centralized funding bodies that must assess applications and partition the resources to contribute to a number of priorities. Primary examples of such institutions are the National Institutes of Health (NIH), the National Science Foundation (NSF) and the Department of Defense (DOD). Each of these institutions makes many details of their procedures publicly available, which we describe in Section 3. We connect their allocation problems to established priority-design schemes: earmarks and floors. *Earmarks*, commonly termed *reserve systems* in the literature, have been proposed and implemented primarily to allocate discrete resources, and often for affirmative-action purposes. These work by setting aside portions of the resource equipped with separate priority orders to ensure adequate consideration of each type of recipient while mitigating justified envy: never assigning a resource to a claimant who is lower priority than someone who did not receive it. Alternatively, type-specific *floors*, or minimum targets, pursue the same goal without partitioning the resource: each type is targeted to receive at least a given amount, and beyond that, all claimants compete under a single priority order. We show how the same two approaches can be adapted to balance several social priorities when a continuous budget is divided among projects that differ in cost and may be partially funded. In doing so, we introduce an important new application to the priority-design literature.

In our model, each applicant reports two numbers, the amount of funding it asks for (the grant's maximum cost) and the least it would accept (its minimum cost), together with the funding categories it is eligible for. Given the report, peer review determines a merit score. The combination of the report and the merit score is called a *grant*, which is what the designer observes; they can choose to reject a grant, or fund it at any level between its minimum and maximum cost. Each category carries a constraint, an earmark or a floor, and these sum to at most the total budget. When the constraints are earmarks, requiring, as Delacrétaz (2021) does, that categories split the cost of a grant evenly yields a unique characterization: a rule is non-wasteful, respects priorities, and is eligibility-compliant and category-neutral if and only if it is the *serial-simultaneous reserve (SR) choice rule* (Theorem 1). Moreover, the serial-SR choice rule is *monotonic in types*, so it never penalizes a grant for a weak improvement: a grant that becomes cheaper, gains merit or claims more categories cannot thereby lose its funding (Proposition 3). Finally, if a grant that is worse than all funded grants is added to the problem, the funding assigned to the existing grants is unaffected; the only possible difference is that leftover funding is assigned to the new grant (Proposition 4). We show the serial-SR choice rule operates in polynomial time (Proposition 5).

When the constraints are floors, we relax them to *feasible floors*: an allocation satisfies feasible floors if, in every category whose floor is unmet, funding the *next-best grant*, the highest-merit eligible grant not yet fully funded, would carry the category past its floor. Our main result is that the *floors-first choice rule* is non-wasteful, respects priorities and satisfies feasible floors, and is also monotonic in cost and independent of non-contending grants (Theorem 2). Monotonicity in cost means that a grant funded at a costlier request would also be funded at a cheaper one, holding its merit and categories fixed. *Independence of non-contending grants* means that adding an application of lower merit than every funded grant, which is not the next-best grant eligible for any of its categories and is too costly to absorb the idle funds, leaves the allocation unchanged. The floors-first choice rule is not monotonic in types, but the failure is narrowly confined. A grant can gain by submitting a worse application only if it is *marginal*, meaning it is the next-best grant eligible for one of its categories (Corollary 1); for every other grant, the rule is monotonic in types (Proposition 6). Since each category has only one next-best grant, at most as many grants as there are categories are exposed, however many apply (Corollary 2), so in a large pool of applicants the exposed share vanishes. The floors-first choice rule also operates in polynomial time (Proposition 7).

We also show that feasible floors are the appropriate relaxation. A stronger notion, *strong feasible floors*, counts a floor as met only when adding funds to any eligible grant would overshoot it. No rule that is non-wasteful, respects priorities and satisfies strong feasible floors is independent of non-contending grants (Proposition 1). The failure is both normative and operational: a cheap, low-merit application can turn a satisfied floor into an unmet one and pull funds away from higher-merit grants, and because whether a floor is met then depends on the cheapest eligible grant rather than the next-best one, the funder has no clear margin at which to decide.

Since the floors-first choice rule does not characterize all choice rules that are non-wasteful, respect priorities and satisfy feasible floors, we turn to a refinement of this set of choice rules. We focus on *integer floor problems*, floor problems whose numeric values are all integers, and *integer-valued choice rules*, choice rules that are defined on integer floor problems and assign integer values of funding. The floors-first choice rule is an integer-valued choice rule when restricted to integer floor problems (Proposition 9). The *merit-maximal choice rule* selects, among the integer allocations satisfying the axioms, the one that gives the most funding to the highest-merit grant, then, among those, the most to the second-highest-merit grant, and so on. In other words, it is constrained lexicographically efficient. It is unique (Proposition 10), but it fails independence of non-contending grants, it is not monotonic in cost, merit, or categories, and it is non-monotonic in types for non-marginal grants (Theorem 3). Therefore, the merit-maximal choice rule fails the incentive criteria along all dimensions, and it is sensitive to the addition of grants which should not be in contention

for funding. We read this as evidence that, for floor problems, it is better to give up some efficiency, which supports the floors-first choice rule.

The chief contribution of this paper is to extend priority design and reserve design to grant funding, and more broadly to any setting in which funds are allocated to meet several social objectives. Whereas prior work has largely focused on the allocation of discrete resources, we study the allocation of a pool of funds under the two schemes institutions use in practice. As such, the framework has relevance to an array of applications, such as determining financial aid and scholarships, government budgeting, and disaster relief triage.

The paper proceeds as follows. Section 2 reviews the related literature. Section 3 describes the review and funding procedures of the DOD, NSF and NIH, which motivate the two constraint schemes we study. Section 4 establishes the formal model and desiderata. Section 5 treats earmark problems, Section 6 treats floor problems, and Section 7 studies the merit-maximal choice rule for integer floor problems and shows that its efficiency comes at the cost of the incentive and independence properties satisfied by the floors-first choice rule. Section 8 concludes. Proofs are in Appendix A.

## 2 Related Literature

This paper considers two possible schemes to ensure resources are allocated (or matched) to recipients in a way that respects priorities, while also meeting certain requirements on the distribution of outcomes. There is a deep literature studying this type of question in the context of the college admissions problem and related matching problems which provides an array of approaches to achieve this goal. Abdulkadiroğlu (2005) and Abdulkadiroğlu and Sönmez (2003) introduce type-specific upper quotas or ceilings to the college admissions problem. The former paper considers the college admissions problem, while the latter introduces this approach to school choice. Since then, a number of papers consider different schemes, including reserves, floors, and quotas, to ensure prioritization of each category of recipient in a matching framework.

A long-standing observation about priority-design problems is that allocation decisions are not and should not be determined by lining up candidates according to merit and picking the top $n$ candidates. Rather, a number of distributional objectives are taken into consideration to make a balanced selection (Alkan and Gale 2003, Abdulkadiroğlu 2005). Likewise, in a grant-funding problem, there is a merit-score determined by peer review, and funding decisions are made with consideration of this score as well as institutional objectives to create a balanced portfolio of grants. We refer to this process—peer review, then portfolio balance—as "two-tiered" review, following the convention from the Department of Defense (Department of Defense, 2023).

Hafalir, Yenmez, and Yildirim (2013) study the school choice problem when there are majority quotas versus minority reserves. They are the first to introduce minimum-guarantee reserves, where at each school, students belonging to the affirmative-action category get priority over a portion of school seats until enough students of the category receive a seat; after that point, the seats have a priority order that is not affected by category-inclusion. They find that minority reserves produce an outcome that Pareto-dominates the outcome under majority quotas. They also find that minority reserves produce more favorable outcomes in a simulation exercise. We do not study soft reserves. We treat hard (set-aside) reserves and a feasible version of hard floors separately, which lets us focus on what is new in a grant-funding problem.

Ehlers, Hafalir, Yenmez, and Yildirim (2014) consider the school choice problem with type-specific lower and upper bounds (floors and ceilings, respectively) at each school. They consider an assignment feasible if it assigns every student a seat and it meets the floors and ceilings. They show that, under these hard bounds, an assignment that is fair across types need not exist, and neither need one that is fair for same types and non-wasteful. They propose using soft quotas instead, and they propose the *student exchange algorithm*, which is fair for same types and constrained non-wasteful as well as constrained efficient. A notable feature of their choice rule is that, at each school, it dynamically gives high priority for students belonging to a type that has not reached the floor, medium priority when the floor is met but the ceiling is not exceeded, and low priority when the ceiling is exceeded. Our floors-first choice rule works in a similar way: it gives priority to unmet floors until they are satisfied, and then assigns the remaining funds by merit. As in their setting, hard floors cannot always be met, so we relax them to feasible floors; the floors-first choice rule then respects merit priorities and satisfies feasible floors while remaining non-wasteful (Theorem 2).

A number of papers have studied matching with distributional constraints such as floors and ceilings for each category of student. Kamada and Kojima (2015) study the regional caps Japan places on the number of medical residents matched to each region. The mechanism used in practice meets the caps by lowering the capacities of individual hospitals and then running deferred acceptance (DA); they propose a mechanism that meets the same caps more efficiently. Fragiadakis and Troyan (2017) discuss an improvement on previous solutions to this type of matching problem. They propose a mechanism that Pareto-dominates artificial caps deferred acceptance (ACDA), their name for meeting distributional constraints by lowering capacities and then running DA. Namely, the improved algorithm works by iteratively lowering the overall quotas and type-specific quotas according to a sequence, each time checking whether floors and quotas are met—if so, the procedure stops.

Axiomatic methods entered school choice at the level of the mechanism: Kojima

and Ünver (2014) characterize the Boston mechanism by axioms that include conditions on priorities. Echenique and Yenmez (2015) are the first to apply this approach to a school's choice rule under diversity constraints, showing how real-world constraints pin down a unique choice rule. For earmark problems we give an axiomatic characterization. For floor problems the axioms do not single out a rule; we construct one that satisfies them, and we use a total order to select the maximum of the set of allocations satisfying them, which we show is nonempty.

Later, a few papers discuss school-choice problems where students may be eligible for multiple categories. For instance, Kurata, Hamada, Iwasaki, and Yokoo (2017) are the first to give treatment to this problem, while Sönmez and Yenmez (2019b) follow by considering two possible treatments of "overlapping reserves." In the school choice context, this would mean that they consider a case where students are eligible for multiple reserve categories. They consider two possible treatments, one-to-one and one-to-all. In the first case, a student will satisfy only one reserve slot, and in the latter case, a student satisfies reserve slots for all of her eligible categories. We make use of the one-to-one approach for earmark problems, and the one-to-all approach for floor problems.

Dur, Kominers, Pathak, and Sönmez (2018) show the significance of the order in which reserves are processed. Since then, some more recent papers have considered choice rules that do not depend on an ordering of categories, for instance Delacrétaz (2021). The serial-SR rule does not depend on an order of the categories, and the floors-first rule depends on one only through its ordered partition, which may consist of a single group.

More recent papers have explored additional applications of matching theory, especially related to triage, using schemes that build on Hafalir et al. (2013). For instance, Pathak, Sönmez, Ünver, and Yenmez (2024) consider the allocation of antiviral treatment and other scarce medical resources. They propose a reserve-system approach. In addition, Sönmez and Yenmez (2022) use matching theory to describe the court-mandated job assignment rule for public jobs in India with an axiomatic characterization. Sönmez and Yenmez (2019a), an earlier version of Sönmez and Yenmez (2024), implicitly use a merit total-order to characterize a unique optimal choice rule. The published version was rewritten and no longer contains this construction; we adopt the idea and give it an explicit treatment in our setting. Delacrétaz, Kominers, and Teytelboym (2023) apply matching theory to refugee resettlement, where families of differing sizes create knapsack-type complementarities at localities. Because families differ in size, a stable matching need not exist, and finding one is computationally intractable. Envy-freeness can always be met but, under multidimensional constraints, it forbids priority violations that harm no one and so generates unnecessary waste. They therefore introduce *interference-freeness*: a family may be envied provided that no higher-priority family that weakly prefers its locality could

claim any of the capacity it uses. Interference-freeness is implied by envy-freeness, is logically independent of stability, and is attained in polynomial time by a variant of deferred acceptance, at the cost of tolerating some waste. Our floor axioms follow the same logic. Strict floors, like stability, can fail to exist because of an indivisibility, here a grant's minimum cost (Section 4.4). Strong feasible floors (Axiom C$^*$) play the role of envy-freeness: they test a shortfall against every eligible grant, including grants far below the merit frontier, so the arrival of a cheap low-merit application can change the allocation (Proposition 1). Feasible floors (Axiom C) play the role of interference-freeness: they test the shortfall only against the claim the priority structure puts next in line, the highest-merit eligible grant not yet fully funded, and the floors-first algorithm attains them in polynomial time (Theorem 2). Two differences are worth noting. Floors are distributional targets rather than priority constraints, and unlike interference-freeness, feasible floors are compatible with non-wastefulness. The closer formal counterpart of interference-freeness in our setting is the first clause of respecting priorities (Axiom B), which counts a priority violation only when the funds held by lower-priority grants, together with any unspent budget, could actually cover the higher-priority grant. Like these applications, our paper extends matching theory to a setting with complementarities (induced by funding floors), where traditional matroid structures no longer apply.

Biró, Fleiner, Irving, and Manlove (2010) study higher education in Hungary and demonstrate the computational difficulty of matching with upper and lower capacity constraints for schools: they find that determining whether a stable matching exists for this type of problem is NP-complete. In our model, by analogy, each grant has a lower and an upper demand, its minimum and maximum cost, while the institution faces only an upper capacity, its budget.

Kesten and Ünver (2015) are the first to study school choice mechanisms that determine lotteries over assignments, which is related to our study of fractional assignment. While student-proposing DA with random tie-breaking is ex-post stable for the school choice problem (Abdulkadiroğlu & Sönmez, 2003), Kesten and Ünver study broader ex-ante notions of fairness for stochastic mechanisms. They find a choice rule leading to *strongly ex-ante stable* matching lotteries. That is, a student may have a positive probability of a school only if every student with higher priority there has no probability of a school she likes less; and of two students with equal priority at a school, one may have a higher probability of it only if the other has no probability of a school she likes less. Their first proposed choice rule *fractional deferred acceptance* (FDA) works along the lines of DA, but when equal priority students apply to the same school, if there is not enough capacity, the capacity will generally be shared as an equal assignment probability between the top-priority students. The second choice rule they study, *fractional deferred acceptance with trading* (FDAT) begins with the FDA outcome, and uses *ex-ante stable improvement cycles* much in the same fash-

ion as the top trading cycle algorithm (Shapley & Scarf, 1974), finding a cycle of students who each prefer another student's assignment fraction, and trading these fractions accordingly.

Delacrétaz (2021) studies a reserve design context where a social planner has set aside reserves for a number of uses, or categories. Each recipient demands at most one unit, and Delacrétaz proposes drawing that unit evenly from the categories for which the recipient is eligible, a desideratum he calls *category-neutrality*. He shows that the simultaneous reserve allocation is eligibility-compliant, non-wasteful, respects priorities and is category-neutral, and that every allocation with these four properties gives each recipient the same total amount. The characterization is therefore of the aggregate allocation: when categories rank recipients differently, distinct allocations satisfying all four properties can split the same totals differently across categories. Delacrétaz notes that the *simultaneous reserve algorithm* can be read as deferred acceptance in which categories propose to recipients and, up to small differences in setup, is equivalent to the fractional deferred acceptance algorithm of Kesten and Ünver (2015). Unlike Kesten and Ünver, who consider many schools but no category-specific reserves, Delacrétaz considers multiple reserve categories and one institution; hence it is closer to the grant-funding context we consider. In a discrete matching context, the outcome for each agent may be interpreted as the probability of assignment. However, in our context, we take advantage of the fact that budgets are continuous, so a similar rule can be used to determine exact funding amounts assigned from reserve categories to grants. Hence, the section we dedicate to earmarks is in the same spirit as Delacrétaz (2021). Our context is less general in that a category's priority order reflects the overall merit order when comparing grants that are both eligible, as in Sönmez and Yenmez (2019b). This enables a simpler solution: a single pass through the merit order that settles each grant at its own step. In Delacrétaz (2021), the basic *simultaneous reserve algorithm* reallocates capacity across rounds and can cycle without terminating, reaching the simultaneous reserve allocation only in the limit; a variant that solves a linear program between rounds finds it after finitely many rounds, polynomially many in the numbers of recipients and categories. Our model is distinct in that grants are allowed to have arbitrary minimum and maximum costs, an important detail if grants are meant to be heterogeneous and reflect projects of different scales.

A large literature relaxes strategy-proofness in matching markets, for instance by requiring it only in the limit as the market grows (Azevedo & Budish, 2019; Kojima & Pathak, 2009). Our incentive criteria are of a different kind. Strategy-proofness compares a truthful report with a misreport and evaluates the comparison by the agent's own preferences; as we argue in Section 4.7, a grant application has no truthful counterpart, since the applicant authors the project it proposes. We instead compare two applications the same investigator might submit, ordered by

the institution's partial order over types, and require that the rule never fund the worse one while rejecting the better. The closest antecedent is the requirement that a mechanism *respect improvements*, introduced by Balinski and Sönmez (1999) for student placement: a student's assignment should not deteriorate when her test scores improve. Monotonicity in merit is the analogue of that condition in our setting, and monotonicity in types extends it to the dimensions an applicant controls (cost bounds and claimed categories). The condition is also familiar from mechanism design with transfers, where a monotone allocation rule, as in the auction of Vickrey (1961), is what makes truthful reporting implementable; absent transfers, monotonicity is the part of that requirement that remains meaningful. We also consider monotonicity along each dimension separately when the full criterion fails.

Finally, this paper builds on the established insight that incentive requirements place limitations on other desirable properties of a mechanism (Theorem 3). Since this tradeoff is widespread in the literature, we cannot mention all papers that make this observation, and focus on two important examples. First, Roth (1982) finds a tradeoff between stability and incentives for truth-telling for all agents in two-sided matching mechanisms (the marriage problem). This observation has been applied to the school choice context: Pathak and Sönmez (2008) compare the Boston mechanism, which is manipulable, with deferred acceptance, which is strategy-proof. They find that the Boston mechanism rewards sophisticated agents at the expense of sincere ones, so the strategy-proof mechanism may be considered more fair, as it does not disadvantage less sophisticated agents. For floor problems, we show that among integer-valued choice rules that are non-wasteful, respect priorities and satisfy feasible floors (Axioms A–C), no rule that is monotonic in a single dimension of the type (cost, merit, or categories), or monotonic in types for non-marginal grants, is merit-maximal (Theorem 3).

# 3 Two-Tiered Review

## 3.1 Department of Defense (DOD)

We consider three grant-making institutions that follow the same broad pattern in making funding decisions: (1) they review applications to assess their merit, and (2) in light of these assessments and the programmatic categories each application claims, they choose a balanced portfolio of grants. First, the Department of Defense funds the Congressionally Directed Medical Research Programs, which gives the following description of their process, "To ensure both scientific excellence and programmatic relevance, the CDMRP administers a two-tiered review process consisting of peer and programmatic review" (Department of Defense, 2023). We include an infographic as

Figure 1 found on the same webpage, which illustrates their process.

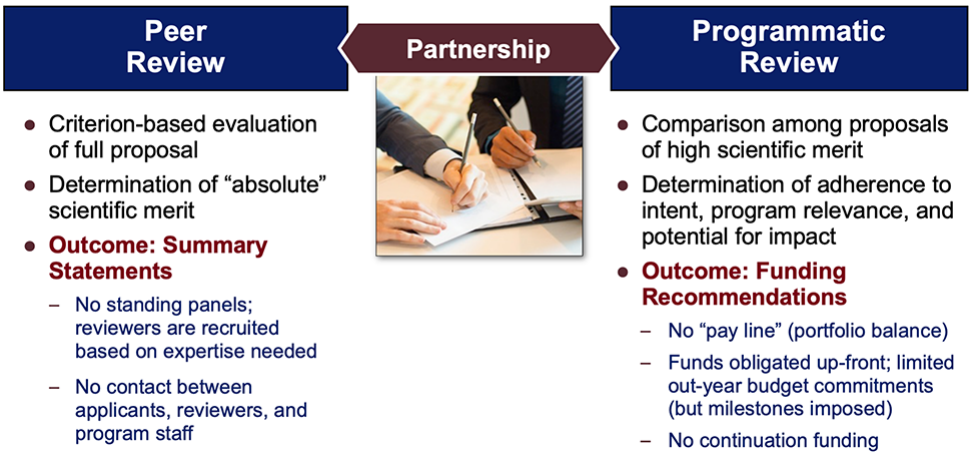



Figure 1: Two-Tiered Review

## 3.2 National Science Foundation (NSF)

The NSF lists its goals for grant funding, and provides the following as the first bullet, "Maintaining a balanced portfolio that provides opportunities for original research in all fields of science, engineering, and learning" (National Science Foundation, 2022, p. 31). Like the DOD, the NSF uses the terminology of a "balanced portfolio" as a goal for the second tier of review, which means merit and potential impact alone do not determine funding decisions.

## 3.3 National Institutes of Health (NIH)

Likewise, the NIH provides clear details about their process, describing the first stage as peer review and the second as a review that weighs research priorities and the balance of the existing portfolio[1]:

> Funding decisions are made through a rigorous dual-level review process. Scientific review groups or study sections, first evaluate and score research proposals for scientific and technical merit. Study sections are made up of volunteer scientists, mostly from academia, and overseen by NIH staff known as scientific review officers.
>
> Advisory councils for NIH ICs and the NIH Office of the Director then perform a second-level review for mission relevance. Ultimately, IC directors

[1]We provide a more detailed description of the NIH's review and funding policies in Appendix B.

> make final funding decisions, taking into consideration current research priorities and the existing funding portfolio. (Talesnik, 2025)

Additional sources concur with this description of the NIH peer review process (NIH, 2021). That the second stage matters is also visible in the data: at the National Institute of Neurological Disorders and Stroke, a substantial share of funding goes to grants scored below the "payline," so many funded grants have lower merit than some rejected ones (National Institute of Neurological Disorders and Stroke, 2021).

Our model abstracts from the details of these processes and keeps their common structure. The first tier is summarized by a single strict merit order over applications. This is a modeling assumption rather than a feature the sources above establish: review produces scores, which may tie and which some applications never receive, and we take the merit order to be a strict ranking that extends them (see the discussion following Theorem 1). The second tier is summarized by a set of categories, each standing for a programmatic objective, and an application may be eligible for several of them. An application's categories are those it claims when it is submitted; peer review determines merit but not category membership. Where a category reflects a verifiable status, such as career stage, the claim is simply checked. We consider two ways a policymaker might pursue its portfolio objectives. Under earmarks, part of the budget is set aside for each category and may be spent only on grants eligible for it. Under floors, each category has a target minimum level of funding for its eligible grants. In both cases, we impose two core requirements (Section 4.3): usable funds should not sit idle, and no grant should be funded ahead of one with a stronger claim to the same funds. Merit enters through the second.

# 4 Setup

## 4.1 Earmark Problems

The first type of grant-funding problem we consider is an earmark problem, where funding is sectioned off into reserves whose use is strictly limited to eligible grants.

There is a finite set $I$ of grants applying for funding and a total budget $\mathcal{B} \in \mathbb{R}_+$. The budget is divided among a finite set of categories $\mathscr{C}$, made up of an open category $o$ and the reserved categories $\mathscr{C} \setminus \{o\}$: each category $c$ is given a *reserve* $q_c \in \mathbb{R}_+$, with $\sum_{c \in \mathscr{C}} q_c = \mathcal{B}$. Each grant $i \in I$ has a *minimum cost* $\underline{b}_i$ and a *maximum cost* $\bar{b}_i$, the amount it requests, with $[\underline{b}_i, \bar{b}_i] \subseteq (0, \mathcal{B}]$; we call these its *cost bounds*, and say that $i$ is *fully funded* when it receives $\bar{b}_i$. Grant $i$ is *eligible* for each category in $\rho(i) \subseteq \mathscr{C}$, a set that always contains $o$. The set $\rho(i)$ is part of the application: the applicant claims the categories its project addresses, and the institution takes

the claim as given (Section 4.7). Grants are ranked by a strict *merit order* $\succ$, and each category $c$ ranks them by a strict *reserve priority order* $\pi_c$. The open category ranks by merit alone, $\pi_o \equiv \succ$; a reserved category's order agrees with $\succ$ on the grants eligible for it and ignores the rest. In examples we label grants by their merit rank, so that $1 \succ 2 \succ \cdots$, and write $\alpha$ and $\beta$ for categories other than the open category. An *earmark problem* collects these components:

$$\mathcal{G} = \langle I, \mathcal{B}, \mathscr{C}, \succ, \rho, \{[\underline{b}_i, \bar{b}_i]\}_{i \in I}, \{q_c\}_{c \in \mathscr{C}}, \{\pi_c\}_{c \in \mathscr{C}} \rangle.$$

A *disaggregate allocation* $x$ is a nonnegative $|I| \times |\mathscr{C}|$ matrix whose elements $x_{i,c}$ give the assignment from category $c$'s reserve to grant $i$. We denote the $i^{\text{th}}$ row sum $\sum_c x_{i,c}$, the total funding to grant $i$, as $x_i$, and the $c^{\text{th}}$ column sum $\sum_i x_{i,c}$, the total drawn from category $c$'s reserve, as $x_c$. The matrix must satisfy either $x_i = 0$ or $\underline{b}_i \leq x_i \leq \bar{b}_i$ for all grants $i$, and $0 \leq x_c \leq q_c$ for all categories $c$. If a matrix $x$ fails one of these conditions, we say $x$ is *unattainable*. An *allocation* $\xi \in \mathbb{R}_+^{|I|}$ is a vector of row sums of a disaggregate allocation. Hence, an allocation gives the total amount of funding assigned to each grant. An allocation is *attainable* if for all $i$, either $\xi_i = 0$ or $\underline{b}_i \leq \xi_i \leq \bar{b}_i$. A *disaggregate choice rule* $\hat{C}$ maps each earmark problem to a disaggregate allocation, $\hat{C} : \mathcal{G} \mapsto x$, and a *choice rule* $C$ maps each earmark problem to an allocation, $C : \mathcal{G} \mapsto \xi$.

## 4.2 Floor Problems

A policymaker may want more flexible use of the budget. In a *floor problem*, $q_c$ is no longer a reserve but a *floor*: the approximate minimum amount that should go to grants eligible for category $c$. Floors need not exhaust the budget, so we drop the open category $o$ and require only $\sum_{c \in \mathscr{C}} q_c \leq \mathcal{B}$; since no category holds funds of its own, we also drop the reserve priority orders $\pi_c$. A floor problem collects the remaining components:

$$\mathcal{G} = \langle I, \mathcal{B}, \mathscr{C}, \succ, \rho, \{[\underline{b}_i, \bar{b}_i]\}_{i \in I}, \{q_c\}_{c \in \mathscr{C}} \rangle.$$

Because floors do not assign funds to categories, there is no analogue of a disaggregate allocation. An *allocation* is a vector $\xi \in \mathbb{R}_+^{|I|}$ of funding amounts, and it is *attainable* if it meets the per-grant condition of Section 4.1 and $\sum_{i \in I} \xi_i \leq \mathcal{B}$. For a category $c$, let $I_c = \{i \in I : c \in \rho(i)\}$ be the grants eligible for $c$, and let

$$A_c(\xi) = \{i \in I_c : \ \xi_i < \bar{b}_i\}, \qquad d_c(\xi) = q_c - \sum_{i \in I_c} \xi_i.$$

Thus $A_c(\xi)$ is the set of grants eligible for $c$ that are not fully funded at $\xi$, and $d_c(\xi)$ is the gap between the floor and the funding $\xi$ assigns to grants eligible for $c$. The

gap is negative when that funding exceeds the floor, so the floor is met exactly when $d_c(\xi) \leq 0$; when $d_c(\xi) > 0$ we call it the category's *shortfall*. A *choice rule* $C$ maps each floor problem to an allocation, $C : \mathcal{G} \mapsto \xi$. A *grant-funding problem* is either an earmark problem or a floor problem.

## 4.3 Core Axioms

We develop a set of axioms providing the desiderata for each type of problem. We will say a choice rule $C$ satisfies an axiom if it always leads to allocations satisfying the axiom. Axioms A–C define the acceptable allocations in floor problems, and Axioms A, B, D and E characterize the solution to earmark problems, so Axioms A and B apply to both.

**Axiom A.** *An allocation $\xi$ is **non-wasteful** if there is no attainable $\xi' \neq \xi$ with $\xi' \geq \xi$. In an earmark problem the requirement is placed on the disaggregate allocation, and the comparison ranges only over matrices that respect the designation of funds: there is no attainable $x' \neq x$ with $x' \geq x$ and $x'_{i,c} = 0$ for every grant $i$ and every $c \notin \rho(i)$.*

Non-wastefulness can be checked one grant at a time. Write

$$s(\xi) = \mathcal{B} - \sum_{j \in I} \xi_j \qquad \text{and} \qquad r_i(x) = \sum_{c \in \rho(i)} \big(q_c - x_c\big)$$

for the funds that remain available to grant $i$: the unspent budget in a floor problem, and the unassigned reserves in $i$'s own categories in an earmark problem. Then $\xi$ is non-wasteful if and only if, for every $i \in I$,

$$\xi_i = 0 \implies s(\xi) < \underline{b}_i \qquad \text{and} \qquad \underline{b}_i \leq \xi_i < \bar{b}_i \implies s(\xi) = 0, \tag{1}$$

and a disaggregate allocation $x$ with $x_{i,c} = 0$ for every $c \notin \rho(i)$ is non-wasteful if and only if (1) holds with $x_i$ and $r_i(x)$ in place of $\xi_i$ and $s(\xi)$.[2] That is, an unfunded grant is passed over only if what remains available to it is below $\underline{b}_i$, and a funded grant stops short of $\bar{b}_i$ only if nothing remains.

The earmark clause compares $x$ only with matrices that respect designation, since an expansion drawing on a category a grant is not eligible for is not one the institution may carry out. Axiom B below counts available funds the same way, through $s(\xi)$ and $r_i(x)$. This is the *constrained* form of non-wastefulness, as in Ehlers et al. (2014).

[2] If an attainable $x' \geq x$ differs from $x$, pick a grant $i$ with $x'_i > x_i$ and let $x''$ agree with $x'$ in row $i$ and with $x$ elsewhere. Every column sum of $x''$ is at most that of $x'$, so $x''$ is attainable, respects designation, and expands $x$ in row $i$ alone. Such an expansion exists exactly when $r_i(x) \geq \underline{b}_i$ if $x_i = 0$, and when $r_i(x) > 0$ if $\underline{b}_i \leq x_i < \bar{b}_i$. The floor case is identical with $s(\xi)$ in place of $r_i(x)$.

Priority combines merit with categories. In an earmark problem, each reserve's priority order $\pi_c$ says which grants have the better claim to it. A floor problem has no reserves, so we compare grants directly, using the *priority partial order* $\succ_\rho$: $i \succ_\rho j$ if $i$ has strictly higher merit than $j$ and is eligible for every category $j$ is eligible for.

**Axiom B.** *In a floor problem, an allocation $\xi$* ***respects priorities*** *if*

1. *for every $i \in I$ with $\xi_i = 0$, there is no nonempty $J \subseteq \{j \in I : \xi_j > 0$ and $i \succ_\rho j\}$ with $\sum_{j \in J} \xi_j + s(\xi) \geq \underline{b}_i$; and*

2. *for every $i \in I$ with $\underline{b}_i \leq \xi_i < \bar{b}_i$, there is no $j \in I$ with $i \succ_\rho j$ and $\xi_j > 0$.*

*In an earmark problem the same requirement is imposed reserve by reserve: a disaggregate allocation $x$ respects priorities if*

1. *for every $i \in I$ with $x_i = 0$, there is no $J \subseteq I$ containing some $j$ with $x_{j,c} > 0$ for a category $c$ having $i \,\pi_c\, j$, such that $\sum_{j \in J} \sum_{c \,:\, i \,\pi_c\, j} x_{j,c} + \sum_{c \in \rho(i)} \big(q_c - x_c\big) \geq \underline{b}_i$; and*

2. *for every $i \in I$ with $\underline{b}_i \leq x_i < \bar{b}_i$, there is no $j \in I$ with $i \succ j$ and $\sum_{c \in \rho(i)} x_{j,c} > 0$.*

Each clause forbids one kind of priority violation. The first says that no collection of funds going to grants with a worse claim, together with the funds still available to $i$ (the unspent budget in a floor problem, the reserves still standing in $i$'s own categories in an earmark problem), could have covered $i$ instead. The second says that once a grant with a better claim has been funded but not fully funded, nothing should go to a grant with a worse claim. This is the counterpart of "eliminates justified envy" in the matching literature, with the qualification that merit alone does not settle who has the better claim: a grant of higher merit that is eligible for fewer categories is not comparable under $\succ_\rho$, and neither grant has a claim against the other.

Both versions of the first clause carry a proviso: a grant with a worse claim must hold funds that $i$ could use. In a floor problem this means $J$ is nonempty; in an earmark problem, $J$ must contain some grant $j$ holding funds from a category $c$ with $i \,\pi_c\, j$. Without the proviso, $J = \emptyset$ would be admissible, and the clause would rule out every allocation in which the funds available to $i$, $s(\xi)$ or $\sum_{c \in \rho(i)} (q_c - x_c)$, are by themselves at least $\underline{b}_i$. Axiom A already rules these out, since $i$ could be funded from those funds without taking anything from anyone, and no grant with a worse claim is involved. The proviso ensures that every violation of the clause involves a grant with a worse claim holding funds $i$ could have used; once such a grant is present, funds that are still available to $i$ count toward covering $i$, since they could be combined with what the worse claims release.

Non-wastefulness and respecting priorities are the universal baseline for both earmark and floor problems. Any acceptable allocation should make full use of the funds at hand, and it should respect the priority order.

## 4.4 Floor-Specific Axioms

Quotas, upper bounds on what a category's grants may receive, could also help balance a portfolio, but not when grants are eligible for several categories. If a grant $i$ is eligible for categories $c$ and $c'$, it would be undesirable to withhold funding from $i$ because $c$ has reached its quota while $c'$ is still short of its floor. We therefore study floors alone.

There is one caveat to implementing floors: because each grant has a minimum cost, floors cannot always be met exactly. For example, there may be three grants costing 4 each, and a budget of 10. If the first two grants are eligible for category $\alpha$ and the last is eligible for category $\beta$, and floors are 5 for each of $\alpha$ and $\beta$, there is no way to meet floors; either 8 goes towards category $\alpha$ and zero towards $\beta$, or 4 goes to each. We therefore need a relaxation of floors that can always be met, which we call *feasible floors*.

**Axiom C.** *An allocation $\xi$ **satisfies feasible floors** if for every category $c \in \mathscr{C}$ with $d_c(\xi) > 0$, the highest-merit $i^* \in A_c(\xi)$ satisfies $\xi_{i^*} = 0$ and $\underline{b}_{i^*} > d_c(\xi)$.*

**Axiom C$^*$.** *An allocation $\xi$ **satisfies strong feasible floors** if for every category $c \in \mathscr{C}$ with $d_c(\xi) > 0$ and every $i \in A_c(\xi)$, $\xi_i = 0$ and $\underline{b}_i > d_c(\xi)$.*

Axiom C$^*$ requires that whenever category $c$ falls short of its floor, every eligible grant that is not fully funded is unfunded and has a minimum cost exceeding the shortfall $d_c(\xi)$, so that funding any of them would carry $c$ past its floor. Axiom C imposes this requirement only on the highest-merit grant in $A_c(\xi)$, the grant the institution would fund next in category $c$, and is therefore a relaxation of Axiom C$^*$. If every grant eligible for $c$ is fully funded, then $A_c(\xi) = \emptyset$ and both axioms hold vacuously for $c$. We impose Axiom C throughout. The reason, given in Section 4.5, is that the stronger axiom is incompatible, given Axioms A and B, with the independence property introduced there (Proposition 1).

For floor problems, we will find a choice rule that is non-wasteful, respects priorities, and satisfies feasible floors. However, these axioms are insufficient for a unique characterization. We will achieve a unique solution by making use of a total order of allocations induced by the merit of grants in Section 7.

## 4.5 Independence of Non-contending Grants

Fix a floor problem $\mathcal{G}$, let $\xi = C(\mathcal{G})$, and let $j$ satisfy $\xi_j = 0$ (this includes any $j \notin I$). Then $j$ is *non-marginal at* $\xi$ if

1. $\underline{b}_j > s(\xi)$, and

2. for every $c \in \rho(j)$ there is some $i \in I_c$ with $i \succ j$ and $\xi_i < \bar{b}_i$.

Otherwise $j$ is *marginal at* $\xi$. Funded grants are not classified either way: the distinction concerns which unfunded grant the institution would turn to next, so it has no content for a grant already receiving funds.

Say that $j$ is *non-contending at* $\xi$ if it is non-marginal at $\xi$ and, in addition, $i \succ j$ for every $i \in I$ with $\xi_i > 0$, so that $j$ falls below the funded frontier. We state the third requirement separately from the first two because the two notions enter different results. Non-marginality is the restriction under which we compare rules by monotonicity in types on floor problems: the floors-first rule is monotonic in types at every non-marginal grant, whether or not it is non-contending (Proposition 6), while the merit-maximal rule fails monotonicity in types even there (Theorem 3). The property defined below requires, in addition, that a grant be out of contention altogether.

If $j \notin I$, write $\mathcal{G}^{+j}$ for the problem obtained from $\mathcal{G}$ by adjoining $j$ to $I$. A choice rule $C$ is *independent of non-contending grants* if, for every floor problem $\mathcal{G}$ and every application $j \notin I$ that is non-contending at $C(\mathcal{G})$, $C(\mathcal{G}^{+j}) = \big(C(\mathcal{G}), 0\big)$.

In any allocation the institution has exactly two kinds of margin. There is the budget margin, funds sitting unassigned, and there is the merit frontier: the best grant that has not yet been fully funded, which may be considered for each category and overall. Every funding decision the institution can make is a decision at one of these margins: spend idle money, or move down the queue for some category or overall. A non-marginal application sits at neither. It is too large to absorb the idle funds, and in every eligible category there is a better grant with higher priority seeking funding. A non-contending application satisfies one further requirement: it ranks below every grant that received funding, so it could not have displaced any of them. Such an application was never in contention, and its addition should not affect funding decisions.

This property separates the two floor axioms. Axiom C$^*$ quantifies over every eligible grant that is not fully funded rather than the highest-merit one alone, so the binding case is the cheapest such grant, and more funding is required before a floor counts as met. The stronger threshold is incompatible with independence.

**Proposition 1.** *If a choice rule $C$ satisfies non-wastefulness (Axiom A), respects priorities (Axiom B) and satisfies strong feasible floors (Axiom C$^*$), it is not independent of non-contending grants.*

The proof, in Appendix A.4, needs only four grants and a single category. In it, a category's floor counts as met because the next-best eligible grant costs more than the shortfall. A cheap eligible grant of lower merit then arrives. It leaves the next-best grant, and hence Axiom C, unaffected, but under Axiom C$^*$ the floor

becomes unmet, and the only allocation satisfying Axioms A, B and C* defunds the highest-merit grant to fund the newcomer. In general, a threshold set by the cheapest eligible application rather than the best one is disturbed by the arrival of a cheap low-merit grant: a floor previously deemed satisfied now counts as unmet, and closing the gap requires diverting funding into that category and away from higher-merit grants. This represents a normative failure since the addition of low-quality grants alters the allocation. Furthermore, it drastically complicates the problem of finding a suitable choice rule since there is no longer a clear margin for decision-making. Axiom C has neither defect: the floors-first rule of Section 6 satisfies Axioms A–C and is independent of non-contending grants (Theorem 2). This is why we impose Axiom C rather than Axiom C*. The case does not rest on existence: whether Axioms A, B and C* can always be met together is open, and an exhaustive search of small problems, reported in Appendix A.4, finds no counterexample.

## 4.6 Earmark-Specific Axioms

**Axiom D.** *A disaggregate allocation $x$ is* ***eligibility-compliant*** *if $x_{i,c} = 0$ whenever $c \notin \rho(i)$.*

Funding is not used in a way that violates its designation: a grant draws only on the reserves of categories it is eligible for. Since reserve categories have a priority order that prefers to go unassigned over funding a grant outside the category, this axiom agrees with the typical notion of *individual rationality*. Funding is not explicitly split into categories in a floor problem, so the axiom is relevant only to earmark problems.

Axioms A and D are therefore meant to be read together. Eligibility-compliance restricts which reserves may fund a grant. Non-wastefulness compares $x$ only with eligibility-compliant matrices, so it rules out only those expansions that draw on reserves the grants are eligible for. Without that restriction, Axiom A could require expansions that Axiom D forbids, and on some problems no allocation would satisfy both. This happens, for example, when a grant funded below its maximum cost could be topped up from an unassigned reserve it is not eligible for, and no eligibility-compliant allocation can use that reserve up.[3]

Axioms A and B then divide the work without overlapping. Axiom A governs idle designated funds: no grant may absorb more from the reserves of the categories it is eligible for, whether it is unfunded or funded short of its maximum cost. Axiom

[3]For a minimal instance, let $I = \{1\}$ and $\mathscr{C} = \{o, \alpha\}$ with $q_o = 3$ and $q_\alpha = 5$, so $\mathcal{B} = 8$, and let $\rho(1) = \{o\}$ with $[\underline{b}_1, \bar{b}_1] = [1, 8]$. The best eligibility-compliant allocation gives $x_{1,o} = 3$. Yet the matrix with $x_{1,o} = 3$ and $x_{1,\alpha} = 5$ has row sum $8 \in [1, 8]$ and exceeds no reserve, so under an unrestricted comparison that allocation would count as wasteful—as would every other eligibility-compliant one. The restricted comparison is in any case the notion the proofs in Appendix A verify.

B governs funds already committed: nothing may be held by a worse claim while a better one goes unmet, and idle reserve enters its first clause only as a supplement to assigned funds.

**Axiom E.** *Writing $\hat{I}_{i,c}$ for the grants of higher merit than $i$ that are eligible for $c$, a disaggregate allocation $x$ is* ***category-neutral*** *if for every grant $i$ and every category $c \in \rho(i)$,*

$$x_{i,c} < \max_{c' \in \rho(i)} x_{i,c'} \quad \textit{only if} \quad x_{i,c} + \sum_{i' \in \hat{I}_{i,c}} x_{i',c} = q_c.$$

Category-neutrality, which follows Delacrétaz (2021), asks the categories a grant is eligible for to share its cost evenly where they can. The one exception is a category that has run out: $c$ may contribute less to $i$ than another eligible category only if $c$'s reserve is fully allocated to $i$ and the grants of higher merit that are eligible for $c$. For earmark problems, we will show that there is a unique choice rule that is non-wasteful, respects priorities, and is eligibility-compliant and category-neutral.

## 4.7 Type Monotonicity

We treat a grant's cost bounds and claimed categories as choices of the applicant rather than facts the institution could in principle verify. Its merit score is different: review assigns it, and the applicant influences it only through the project it proposes and the categories it claims, a distinction we return to below. Treating the request and the categories as choices is a modeling stance, not a property the institutional descriptions in Section 3 establish, and the argument for it is the following. An applicant does not discover the cost of its project and report it; it decides how large a project to propose, over what range of funding it is willing to carry the work out, and which objectives to claim relevance to. There is accordingly no truthful report against which a submission may be compared, and incentive compatibility in the revelation sense is ill-posed here rather than merely unattainable. The extreme case makes the point: a lone applicant eligible for all categories facing a budget $\mathcal{B}$ may propose a project of any size up to $\mathcal{B}$, and any non-wasteful rule funds it; likewise the highest-priority grant may always request more and, under a rule that respects priorities, receive it. Nothing distinguishes an inflated request from a genuinely larger project, because the applicant authors both. The only check on the size of a request is review: merit scores a project's net value (see below), so a larger request earns a better rank only if the added scope is worth the funds it uses.

What remains is a question about design rather than disclosure. Does the mechanism reward applicants for proposing larger, costlier, or less flexible projects than the work warrants? We approach this comparatively: our criteria rank two applications

the same investigator might submit and require that the rule not favor the one that is worse from the institution's standpoint.

An application presents the institution with a *type* $\theta = (\underline{b}, \bar{b}, R, m)$: cost bounds $[\underline{b}, \bar{b}]$, a set of claimed categories $R \subseteq \mathscr{C}$, and the merit score $m$ returned by peer review. The applicant chooses the first two, and the third follows from the project it proposes. Grant $i$'s type is $\theta_i = (\underline{b}_i, \bar{b}_i, \rho(i), m_i)$, and, keeping with convention, $\theta_{-i}$ denotes the types of all other grants. The set of merit scores $\{m_i\}_{i \in I}$ determines the merit order: $i \succ j$ if and only if $m_i > m_j$, and we assume $m_i \neq m_j$ whenever $i \neq j$. This is the strictness assumption discussed after Theorem 1; actual review scores can tie, and $m_i$ should then be read as a score that also encodes the institution's tie-breaking. Holding the rest of $\mathcal{G}$ fixed, we write $C(\theta, \theta_{-i})$ for the allocation when $i$ submits $\theta$, and say that $C$ *funds* $i$ *at* $\theta$ if $C(\theta, \theta_{-i})_i > 0$.

For types $\theta = (\underline{b}, \bar{b}, R, m)$ and $\theta' = (\underline{b}', \bar{b}', R', m')$, write $\theta \succsim_t \theta'$ if $\underline{b} \leq \underline{b}'$, $\bar{b} \leq \bar{b}'$, $R \supseteq R'$, and $m \geq m'$. This partial order captures when one application is weakly "better" than another. (Notice, this is different from the priority partial order for floor problems $\succ_\rho$, which does not consider cost.)

A choice rule $C$ is *monotonic in types* if for every grant $i$, every $\theta_{-i}$, and every $\theta \succsim_t \theta'$, $C$ funds $i$ at $\theta$ whenever it funds $i$ at $\theta'$. This guarantees that improvements in a grant lead to better outcomes in terms of receiving funding or not. We may also consider monotonicity with respect to each component of the type: $C$ is *monotonic in cost*, *in merit*, or *in categories* if the same holds for all pairs $\theta \succsim_t \theta'$ that differ only in $(\underline{b}, \bar{b})$, only in $m$, or only in $R$, respectively. Finally, $C$ is *monotonic in types for non-marginal grants* if the same holds whenever $i$ is non-marginal at $C(\theta, \theta_{-i})$ (Section 4.5).

These conditions do not reach the applicant in the same way. The cost bounds and the claimed categories are written into the application itself, so monotonicity in cost and in categories concerns how an applicant presents a given project. Merit is not written in: review assigns it, and the applicant affects it only through the project it proposes, the categories it claims, and, via net value, the scale it requests. Monotonicity in merit therefore asks that the rule not overturn the judgment of review, never funding an application it would reject had review thought better of it, so that no applicant gains by proposing weaker work.

### What merit measures

It is worth fixing ideas about what merit measures. Each grant is endowed with some gross value $V_i(\bar{b}_i, \rho(i))$, which depends on its requested funding level and on the categories it claims. The categories enter because they may shape how review values a project, which is one reason an applicant might decline to claim one. Since the deviations we study below vary the cost bounds rather than the categories, we

hold $\rho(i)$ fixed and write $V_i(\bar{b}_i)$. At higher requested levels a project is able to take on more scope and is more likely to provide social value. For instance, more lab materials means more trials are allowed, or more hours of labor means a survey sample may be expanded. It follows that $V_i$ is increasing in $\bar{b}_i$. On the other hand, each dollar assigned to fund project $i$ has an opportunity cost, which we assume is constant $\kappa$. Peer review should therefore rank the merit of grants based upon their net value,

$$m_i = V_i(\bar{b}_i) - \kappa \cdot \bar{b}_i,$$

and $i \succ j$ whenever $m_i > m_j$. Nothing below depends on the form of $V_i$: the order $\succ$ that review induces is a primitive.

Note that $V_i$ may rise at rate $\kappa$ over a range of requested levels, in which case $m_i$ is constant there and an applicant may ask for more while its merit is unchanged.

Merit does not depend on $\underline{b}_i$. Review evaluates the project at the level requested; the minimum cost describes a contingency that review never assesses, and it does not enter the net value above. It matters only at the selection stage, through attainability, through non-wastefulness, and through the threshold at which a category's floor is deemed satisfied (Axiom C).

**Taxonomy of deviations**

An applicant may submit a different application in several ways, and our criteria do not treat them alike. Write $\theta_i$ for the original application and $\theta_i'$ for the alternative. Each row of Table 1 also covers its reverse: undoing a disimprovement is an improvement, and undoing an incomparable deviation is again incomparable.

| Type | Alternative submitted | $\underline{b}_i$ | $\bar{b}_i$ | $m_i$ | vs. $\theta_i$ |
|---|---|---|---|---|---|
| $i$ | Larger request, lower merit | $-$ | $\uparrow$ | $\downarrow$ | disimprovement |
| $ii$ | Larger request, same merit | $-$ | $\uparrow$ | $-$ | disimprovement |
| $iii$ | Larger request, greater merit | $-$ | $\uparrow$ | $\uparrow$ | incomparable |
| $iv$ | Larger minimum cost, same merit | $\uparrow$ | $-$ | $-$ | disimprovement |

Table 1: Deviation Taxonomy

We call a deviation of types *i–iii* a *moonshot*, characterized by an increase in the scale of the project, and hence an increase in the gross value. Types $i$ and $ii$ are *inefficient moonshots*, where the larger scale leads to weakly lower merit; while the gross value rises, it is matched or outpaced by the opportunity cost $\kappa$. In the knife-edge case of type $ii$, the gross value rises at the rate $\kappa$, so merit is unchanged by an expansion in the request. Rewarding expansions that leave merit unchanged makes the planner's problem harder: the larger a grant is relative to the budget, the

more its funding costs the rest of the portfolio. For instance, if a grant is eligible for only a strict subset of the categories, it may be difficult to justify committing an outsized share of the budget to that project; it could pose a challenge to meet the other category floors, or the project could use up the earmarks of all its categories together with the open category's. Therefore, type *ii* is a disimprovement in type. Type *iii* is an *efficient moonshot*, where the larger scale is justified by improved merit. Whether this is an improvement is ambiguous: the grant costs more and so may be harder to fit into the budget, but it offers better value for money. On its own it would be an improvement; as part of a balanced portfolio, it need not be.

Type *iv* is an unambiguous disimprovement: the grant has the same maximum cost and the same merit, but a larger minimum cost. The application is less flexible about the scale at which the work can be carried out, which leaves the institution fewer ways to fit it into the budget, with no gain in merit to offset it. A choice rule should not incentivize this behavior. Two of the disimprovements, types *ii* and *iv*, change only the cost bounds, so monotonicity in cost rules out any gain from them. Type *i* also lowers merit, and monotonicity in types rules out any gain from it.

Our monotonicity conditions are limited in two respects. First, they concern only whether $C$ funds a grant, not how much it receives. An application funded under both proposals, but funded more generously under the worse one, is invisible to them. Type *iv* is where this matters most: a grant cannot secure funding by raising $\underline{b}_i$ under a rule monotonic in types, but it may enlarge an award it would have received in any case. Second, each concerns a grant's own application. A separate requirement, independence of non-contending grants (Section 4.5), concerns the pool rather than the submission: whether an application that was never in contention can change the allocation merely by being submitted. Monotonicity and independence are taken up together when we compare choice rules in Section 7.

# 5 Earmarks

When each category has its own reserve but a grant may be eligible for several categories, how should the reserves fund the grants? Once we require, as Delacrétaz (2021) does, that a grant's eligible categories share its cost evenly where they can, there is only one answer.

We study an earmark problem, which takes the form

$$\mathcal{G} = \langle I, \mathcal{B}, \mathscr{C}, \succ, \rho, \{[\underline{b}_i, \bar{b}_i]\}_{i \in I}, \{q_c\}_{c \in \mathscr{C}}, \{\pi_c\}_{c \in \mathscr{C}} \rangle,$$

where $I$ is a set of grants, $\mathcal{B}$ is the total budget, $\mathscr{C}$ is the set of categories, $\rho$ is the category-eligibility function, $[\underline{b}_i, \bar{b}_i]$ are the cost bounds, $q_c$ gives the earmark for

category $c$, and $\pi_c$ gives category $c$'s priority order. We will find a rule assigning these funds to grants, where the assignment of each pool $q_c$ respects the priority order $\pi_c$.

In words, the disaggregate choice rule satisfying Axioms A, B, D and E (non-wastefulness, respecting priorities, eligibility-compliance, and category-neutrality) considers the grants one at a time in merit order. If the reserves left in a grant's categories can cover its minimum cost, the grant is funded as fully as those reserves and its maximum cost allow, with the cost split as evenly across its categories as they permit, so that the allocation is category-neutral; otherwise the grant is skipped. We illustrate the procedure with an example before stating it formally.

**Example 1.** *Suppose that the total budget is* 10 *and* $q_\alpha = q_\beta = 4$*, so* $q_o = 2$*. Assume grants* $1 \succ 2 \succ 3 \succ 4$ *have cost bounds and categories as follows:*

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $[\underline{b}_i, \bar{b}_i]$ | $[1/2, 1]$ | $[1, 3]$ | $[4, 5]$ | $[3, 4]$ |
| $\rho(i)$ | $\{o, \alpha\}$ | $\{o\}$ | $\{o, \alpha\}$ | $\{o, \alpha, \beta\}$ |

*Then the solution procedure works as follows:*

| | Step 1 | | | Step 2 | | |
|---|---|---|---|---|---|---|
| | $o(2)$ | $\alpha(4)$ | $\beta(4)$ | $o(2)$ | $\alpha(4)$ | $\beta(4)$ |
| 1 | 1/2 | 1/2 | 0 | 1/2 | 1/2 | 0 |
| 2 | 0 | 0 | 0 | 3/2 | 0 | 0 |
| 3 | 0 | 0 | 0 | 0 | 0 | 0 |
| 4 | 0 | 0 | 0 | 0 | 0 | 0 |

| | Step 3 | | | Step 4 | | |
|---|---|---|---|---|---|---|
| | $o(2)$ | $\alpha(4)$ | $\beta(4)$ | $o(2)$ | $\alpha(4)$ | $\beta(4)$ |
| 1 | 1/2 | 1/2 | 0 | 1/2 | 1/2 | 0 |
| 2 | 3/2 | 0 | 0 | 3/2 | 0 | 0 |
| 3 | 0 | 0 | 0 | 0 | 0 | 0 |
| 4 | 0 | 0 | 0 | 0 | 2 | 2 |

The procedure is simple because it is an immediate-acceptance rule: step $t$ changes only the row of grant $i_t$, which is funded, or skipped as grant 3 is in the example, according to the reserves left in its categories alone. We define the *serial-SR disaggregate choice rule* $C^{\text{SR}}$ as the mapping from earmark problems to outcomes $x$ of the serial-simultaneous reserve (SR) algorithm.

List the grants in merit order, $i_1 \succ \cdots \succ i_n$, where $n = |I|$. Let $x^0 = 0 \in \mathbb{R}^{n \times |\mathscr{C}|}$ be the starting matrix of funding, which will be updated one row at a time until $t = n$, in which case let $x$ denote the final result.

**Step** $t \geq 1$**:** Let $g = i_t$ be the grant considered at this step.

For $i \neq g$ and for all $c$, let $x^t_{i,c} = x^{t-1}_{i,c}$. If $\sum_{c \in \rho(g)} (q_c - x^{t-1}_c) < \underline{b}_g$, set $x^t_{g,c} = 0$ for all $c$. Otherwise, set $x^t_{g,c} = 0$ for $c \notin \rho(g)$, and choose $\{x^t_{g,c}\}_{c \in \rho(g)}$ to solve

$$x^t_g = \min\{\bar{b}_g, \sum_{c \in \rho(g)} (q_c - x^{t-1}_c)\} \tag{2}$$

$$x^t_{g,c} = \min\{q_c - x^{t-1}_c, \max_{c' \in \rho(g)} \{x^t_{g,c'}\}\}. \tag{3}$$

**Algorithm 1:** serial-SR algorithm

**Theorem 1.** *A disaggregate choice rule satisfies Axioms A, B, D and E (non-wastefulness, respecting priorities, eligibility-compliance, and category-neutrality) if and only if it is the serial-SR choice rule* $C^{\mathrm{SR}}$.

Two features of Theorem 1 go beyond Delacrétaz (2021). The first is uniqueness of the disaggregate allocation. Delacrétaz characterizes the simultaneous reserve outcome at the level of the aggregate allocation: every allocation satisfying his four axioms gives each recipient the same total, but when two categories rank recipients in opposite orders the axioms admit a continuum of allocations that split those totals differently between the categories. He singles out the simultaneous reserve matrix among these only through an additional criterion, maximal demand (his Proposition 3). Here the four axioms pin down the whole matrix $x$, so they determine not only how much each grant receives but also how its funding is charged to categories. The second is a single-pass procedure. Because every category's priority order is the restriction of one merit order $\succ$, the serial-SR algorithm settles each grant once, at its own step, and halts after $|I|$ of them, each step an explicit computation from the residual reserves (Proposition 5). In Delacrétaz (2021), where categories may rank recipients independently, the basic simultaneous reserve algorithm can cycle without terminating and reaches the allocation only in the limit; finite termination there is obtained by a variant that solves a linear program between rounds. The common merit order is what buys both, and it is a maintained modeling assumption: we take review to deliver a strict ranking of all applications, and a category's priority order to be that ranking restricted to the grants eligible for it. The institutional descriptions in Section 3 and Appendix B establish a first stage of review that scores applications; they do not by themselves deliver a single strict ranking. At the NIH, for instance,

two discussed applications may receive the same overall impact score, and applications that are not discussed receive no score at all. Our results hold for any fixed strict order, so in such an application the order should be read as a strict ranking that extends review's scores to every application. How the institution breaks ties, and where it places unscored applications, is a choice outside the model, and different choices can produce different allocations. The same descriptions also do not establish that every category ranks its eligible applications identically: the second tier weighs programmatic priorities, and we model those priorities through the categories and their reserves rather than through category-specific reorderings of merit. Were categories to rank applications differently, the problem would return to the more general setting of Delacrétaz (2021): the axioms would no longer pin down the disaggregate allocation, and a single pass through the grants would no longer suffice to compute it.

A third feature is that the four axioms are a minimal set for the characterization: drop any one of them and the remaining three no longer single out the serial-SR allocation.

**Proposition 2.** *None of Axioms A, B, D and E is implied by the other three. For each of them there is an earmark problem and an attainable disaggregate allocation that satisfies the other three axioms and fails that one.*

That the serial-SR rule satisfies the four axioms should be intuitive. Eligibility-compliance is enforced by construction, and non-wastefulness is implied because each grant $i$ receives as much funding as the reserves of its eligible categories permit, unless too little remains in those categories to meet the minimum cost $\underline{b}_i$. Since Axiom A compares $x$ only against eligibility-compliant matrices, the reserves standing in categories outside $\rho(i)$ are not funds $i$ was passed over for. Moreover, once the grants of higher merit have been funded, only one amount can go to the next grant without violating Axioms A and B, and category-neutrality fixes how that amount is split among its categories. Finally, each grant's row is settled at its own step and never revisited, so the funding assigned before step $t$ is already the final funding of the grants of higher merit than $i_t$.

Category-neutrality is not free, however. Axiom A tests cellwise expansions of $x$—whether some grant could draw more from a reserve it is eligible for, with every other entry held fixed—and not reallocations of the charges already placed on the categories. When grants have positive minimum costs, Axiom E can leave usable reserves idle that a different eligible split of the same charges would put to use. Example 1 already shows this. Serial-SR spends $1 + 3/2 + 4 = 13/2$ of the budget of 10 and leaves $7/2$ unspent: $3/2$ in $\alpha$ and 2 in $\beta$. Yet the eligibility-compliant allocation that funds grants 1 and 2 at one unit each from $o$, grant 3 at 4 from $\alpha$, and grant 4 at 4 from $\beta$ funds all four grants, exhausts the budget, and satisfies Axioms A, B and D. It fails

only Axiom E, because grant 1 is charged entirely to $o$ while $\alpha$ has reserve to spare. A smaller instance isolates the mechanism.

**Example 2.** *Let $\mathscr{C} = \{o, \alpha\}$ with $q_o = q_\alpha = 1$, so $\mathcal{B} = 2$, and let $1 \succ 2$ with $[\underline{b}_1, \bar{b}_1] = [\underline{b}_2, \bar{b}_2] = [1, 1]$, $\rho(1) = \{o, \alpha\}$ and $\rho(2) = \{o\}$. Serial-SR charges grant 1 half to each category, $x_{1,o} = x_{1,\alpha} = 1/2$. Grant 2 then faces $1/2 < \underline{b}_2$ in $o$ and goes unfunded, and half the budget is left idle. Charging grant 1 entirely to $\alpha$ instead leaves all of $o$ for grant 2, funds both grants, and exhausts the budget.*

Neither outcome violates Axiom A as defined: at $C^{\mathrm{SR}}$ the half unit left in $o$ cannot be given to grant 2, whose minimum cost it does not meet, nor to grant 1, which is fully funded. The loss is therefore not a failure of non-wastefulness but an efficiency cost of Axiom E itself, and it comes from minimum costs. If $\underline{b}_2 \leq 1/2$, the half unit left in $o$ would go to grant 2, as it would in the fractional setting of Delacrétaz (2021), where a recipient can absorb any fraction of a unit. With lumpy grants, splitting the charge of a grant eligible for several categories evenly can leave a residual in a category that is too small for the grants that still need it. Avoiding this requires charging such a grant to the categories that lower-merit applicants cannot use, so that how a grant's funding is split would depend on applications ranked below it, and the resulting rule is not category-neutral; by Proposition 2, Axioms A, B and D alone do not single out an allocation. Category-neutrality should accordingly be read as a substantive choice. It treats the categories symmetrically in how they bear the cost of shared grants, so that no category subsidizes another, and the price is that some designated funds may go unspent that a different apportionment of the same grants would use.

The rule is also well behaved with respect to the applications it receives. Establishing this calls for a different kind of argument, one that compares the allocations produced by two problems differing only in the report of a single grant.

**Proposition 3.** *The serial-SR choice rule $C^{\mathrm{SR}}$ is monotonic in types.*

In other words, the disaggregate choice rule never penalizes a grant for a weak improvement: no deviation that Table 1 classifies as a disimprovement—types *i*, *ii* and *iv*—can secure funding for a grant that would otherwise go unfunded, and neither can claiming fewer categories at the same or lower merit.

Because funding decisions for a grant are determined only by funding towards higher-merit grants, $C^{\mathrm{SR}}$ satisfies a consistency property analogous to independence of non-contending grants, and in fact a stronger one. In an earmark problem the funds available to $j$ are the residual reserves of its own categories rather than the residual budget, so the affordability condition takes the form $\underline{b}_j > \sum_{c \in \rho(j)} (q_c - x_c)$.

**Proposition 4.** *Let $\mathcal{G}$ be an earmark problem, let $x = C^{\mathrm{SR}}(\mathcal{G})$, and let $j \notin I$ satisfy $i \succ j$ for every $i \in I$ with $x_i > 0$. Then $C^{\mathrm{SR}}(\mathcal{G}^{+j})_i = x_i$ for every $i \in I$. If in addition $\underline{b}_j > \sum_{c \in \rho(j)} (q_c - x_c)$, then $C^{\mathrm{SR}}(\mathcal{G}^{+j}) = (x, 0)$.*

The first part says the addition of any grant ranked below every funded grant leaves the incumbent allocation untouched; the second says such a grant is itself unfunded whenever it cannot be covered by the reserves still standing in its categories. Together they give the earmark counterpart of independence of non-contending grants. Note that the first part holds without any affordability requirement at all, which is the sense in which $C^{\mathrm{SR}}$ satisfies more than the floor-problem property asks.

We have shown that fair and non-wasteful allocation of earmarked funds is uniquely determined once eligible categories are required to share the cost evenly, although, as Example 2 illustrates, even sharing can leave usable reserves idle when grants have minimum costs. Further, this unique solution is monotonic in types, so it never funds a weaker application it would otherwise reject (Proposition 3), and it is unaffected by applications that were never in contention (Proposition 4). Floor problems admit no such characterization from our axioms: there, Axioms A–C do not single out a rule.

## 5.1 Computation

The serial-SR algorithm considers each grant exactly once, in merit order, and never revisits a funding decision. All the work within a step consists of splitting one grant's funding among its eligible categories, and the only non-trivial operation that requires is sorting the categories by how much reserve they have left.

**Proposition 5.** *The serial-SR algorithm operates in polynomial time with a total time complexity of* $O(|I| \cdot |\mathscr{C}| \cdot \log(|\mathscr{C}| + 1))$ *and terminates in* $|I|$ *steps.*

# 6 Floors

A floor problem takes the form

$$\mathcal{G} = \langle I, \mathcal{B}, \mathscr{C}, \succ, \rho, \{[\underline{b}_i, \bar{b}_i]\}_{i \in I}, \{q_c\}_{c \in \mathscr{C}} \rangle,$$

where $I$ is a set of grants, $\mathcal{B}$ is the total budget, $\mathscr{C}$ is the set of categories, $\rho$ is the category-eligibility function, $[\underline{b}_i, \bar{b}_i]$ are the cost bounds, and $q_c$ gives the floor for each category, an approximate minimum guarantee. This section constructs a choice rule whose allocations are non-wasteful, respect priorities and satisfy feasible floors.

## 6.1 Algorithmic Construction

A planner who wants to meet every floor, in the sense of feasible floors, while funding the best grants has two natural options, illustrated in Figure 2. In the *sequential* procedure (case 1), the planner takes the categories in a fixed order and, for each

in turn, funds its highest-merit grants until its floor is met. In the *simultaneous* procedure (case 2), the planner repeatedly funds the highest-merit grant eligible for any category whose floor is not yet met, updating which floors are met after each grant. In both, the budget left once every floor is met (the rightmost box) goes to the highest-merit remaining grants regardless of category.

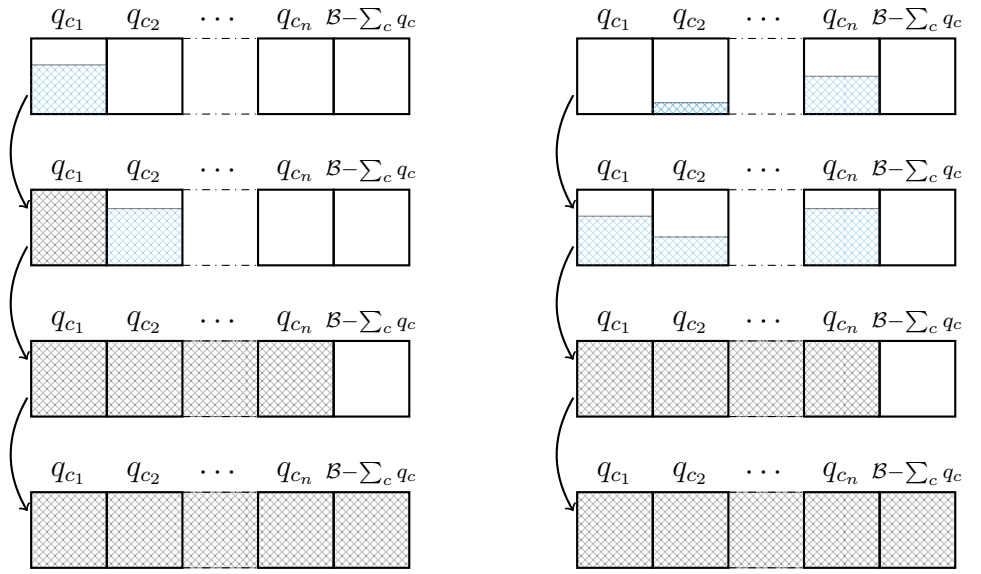



Figure 2: *Two procedures for meeting floors.* Each box is a category's floor $q_c$, and the rightmost box is the remaining budget $\mathcal{B} - \sum_c q_c$. Crosshatching shows funds allocated, and arrows show the order in which funds are assigned.

We will show that either procedure yields a non-wasteful allocation that respects priorities and satisfies feasible floors. Both are special cases of a single procedure. Partition the categories into ordered groups, for example $\{c_1, c_2, c_3\}, \{c_4, c_5\}, \ldots, \{c_{n-1}, c_n\}$, and apply the simultaneous procedure to one group at a time (Figure 3); the remaining budget then goes to the highest-merit remaining grants regardless of category.

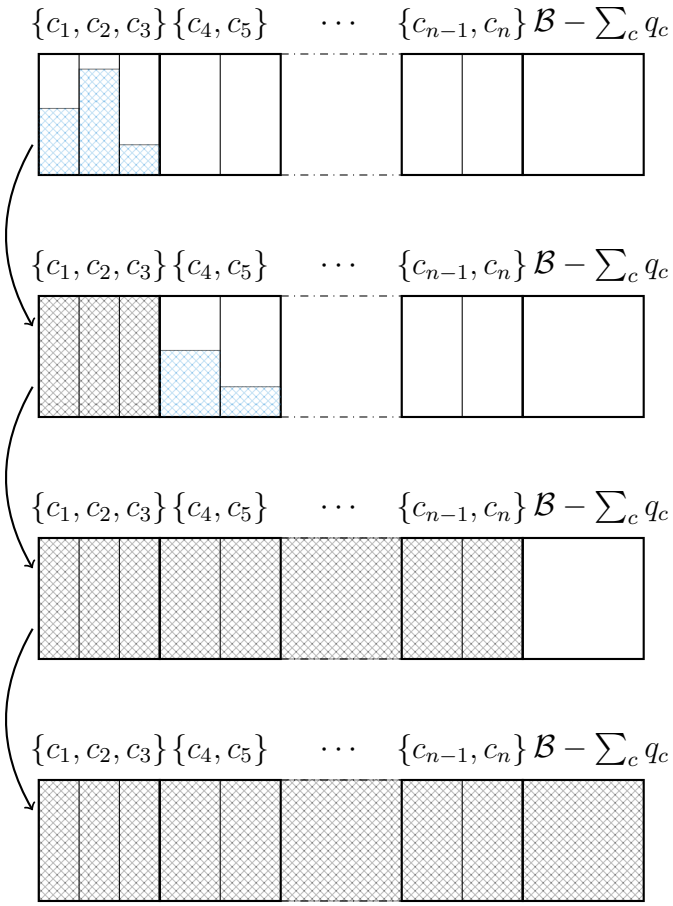



Figure 3: *General procedure.* Categories are partitioned into ordered groups, and the simultaneous procedure is applied to one group at a time before the remaining budget (the rightmost box) is assigned. Crosshatching shows funds allocated.

Groups of size one give the sequential procedure, and a single group gives the simultaneous one. We work with this general procedure for the rest of the section. Since the procedure builds an allocation by adding funding to one grant at a time, we first define the set of grants to which funding can still be added. A grant $i$ is in the *budget set* $F(\xi)$ if either $\xi_i = 0$ and $\underline{b}_i \leq s(\xi)$, or $\underline{b}_i \leq \xi_i < \bar{b}_i$ and $s(\xi) > 0$, where $s(\xi) = \mathcal{B} - \sum_{j \in I} \xi_j$ is the unspent budget (Section 4.3). In other words, $F(\xi)$ contains the grants that can still receive funds at $\xi$: a grant is outside $F(\xi)$ if it is unfunded and $\underline{b}_i$ exceeds the unspent budget, if it is funded short of $\bar{b}_i$ but the whole budget is assigned, or if it is fully funded. The grants in $F(\xi)$ are exactly those that fail

the grant-by-grant condition for non-wastefulness, (1) in Section 4.3, so an attainable allocation $\xi$ is non-wasteful if and only if $F(\xi) = \emptyset$. No grant in $F(\xi)$ is fully funded, so $F(\xi) \cap I_c \subseteq A_c(\xi)$ for each category $c$, but the inclusion can be strict: $A_c(\xi)$ ignores the budget and keeps every grant eligible for $c$ that is not fully funded, including one that the unspent budget cannot pay for. The difference is intended: $F(\xi)$ asks which grants can still be funded from what is left, while Axiom C asks which grants could have been funded had the budget not gone to grants outside the category.

## 6.2 Floors-First Choice Rule

Suppose there is an ordered partition $\Delta = (D_1, \ldots, D_N)$ of the categories $\mathscr{C}$, where the groups $D_k$ are disjoint and have $\bigcup_{k=1}^{N} D_k = \mathscr{C}$. Recall the definition of feasible floors: a category is satisfied if it is not possible to add funding towards the highest-merit grant eligible for the category that is not yet fully funded without exceeding the floor. The rule works through the groups in order. Within the current group $D_k$, it repeatedly funds the highest-merit grant that is not fully funded and is eligible for a category of $D_k$ whose floor is not yet satisfied; once every floor in $D_k$ is satisfied, it moves on to $D_{k+1}$. When all floors are satisfied, it funds the highest-merit remaining grants regardless of category. One feature of this procedure is important for understanding how it behaves: grants may be eligible for multiple categories, and when a grant is funded, it counts towards the floors of *all* categories it is eligible for. This means a later group may have most of its floors met before its turn if earlier steps funded grants eligible for multiple categories. We define the *floors-first choice rule* $C^f_\Delta$ as the mapping from floor problems to the outcome $\xi$ of the floors-first algorithm.

Let $\xi^0 = 0 \in \mathbb{R}^n$ be the initial vector of funding;
**Step** $t \geq 1$:
if $\xi^{t-1}$ fails Axiom C for some category, let $D^t \in \Delta$ be the smallest-index $D \in \Delta$ containing such a category, and define $\mathscr{C}^t \subseteq D^t$ as the set of all categories $c \in D^t$ such that $\xi^{t-1}$ does not satisfy Axiom C for $c$; otherwise, let $\mathscr{C}^t = \emptyset$;
**if** $F(\xi^{t-1}) \neq \emptyset$ **then**
  **if** $\mathscr{C}^t \neq \emptyset$ **then**
    let $g$ be the highest-merit grant in $\bigcup_{c \in \mathscr{C}^t} A_c(\xi^{t-1})$, i.e. the highest-merit grant eligible for some $c \in \mathscr{C}^t$ that is not fully funded; set $M_t = \max_{c \in \mathscr{C}^t \cap \rho(g)} \{d_c(\xi^{t-1})\}$;
  **else**
    let $g$ be the highest-merit grant in $F(\xi^{t-1})$; let $M_t = s(\xi^{t-1})$;
  **end**
  set $\xi_g^t = \xi_g^{t-1} + \min\{\bar{b}_g - \xi_g^{t-1},\ M_t\}$, and $\xi_i^t = \xi_i^{t-1}$ for every $i \neq g$;
  go to the next step;
**else**
  the algorithm terminates, and $\xi \equiv \xi^{t-1}$;
**end**

floors-first algorithm

To summarize, $\mathscr{C}^t$ is the set of categories whose floors are not yet satisfied in the first group $D^t$ that still has one. At each step, the rule chooses the highest-merit grant that is eligible for a category in $\mathscr{C}^t$ and not yet fully funded, and once $\mathscr{C}^t$ is empty, it chooses the highest-merit grant in $F(\xi^{t-1})$, irrespective of categories.

**Theorem 2.** *Fix an ordered partition $\Delta$. The floors-first choice rule $C_\Delta^f$ is monotonic in cost, independent of non-contending grants, and satisfies Axioms A–C (non-wastefulness, respecting priorities, and satisfying feasible floors).*

By Theorem 2, the floors-first choice rule meets every requirement we placed on floor problems: it makes full use of the budget, respects the merit order, and meets every floor in the sense of feasible floors. It is also robust in two respects. A funded grant stays funded if it asks for less, and submitting an application that was never in contention leaves the allocation unchanged. These properties hold for every ordered partition $\Delta$, so they are invariant to the ordering and grouping of the categories, even though the allocation itself may depend on $\Delta$.

The rule's shortcoming is that it is not monotonic in merit or categories. To see why, suppose a grant shrinks its categories or slightly lowers its merit. It may then

no longer complete a category that it previously completed. (A grant *completes* a category if it causes the category's floor to count as met while itself being unfunded, because it is the next-best grant there and is sufficiently expensive.) The rule may then spend more while floors are still unmet, that is, at steps with $\mathscr{C}^t \neq \emptyset$. With less left over for the open phase, a more expensive grant of higher merit may no longer fit, and the deviating grant may be funded in its place. The argument requires the grant to complete a category, which a non-marginal grant never does; indeed, such a grant, unfunded by definition, cannot benefit from a monotonic disimprovement.

**Proposition 6.** *Fix an ordered partition $\Delta$, a floor problem $\mathcal{G}$, and a grant $j \in I$ that is non-marginal at $\xi := C^f_\Delta(\mathcal{G})$. Let $\theta_j \succsim_t \theta'_j$ and let $\mathcal{G}'$ be $\mathcal{G}$ with $j$'s report replaced by $\theta'_j$. Then $C^f_\Delta(\mathcal{G}') = \xi$. In particular $C^f_\Delta$ is monotonic in types for non-marginal grants.*

**Corollary 1.** *Let $\theta_j \succsim_t \theta'_j$. If $C^f_\Delta$ funds $j$ when it reports $\theta'_j$ but not when it reports $\theta_j$, then $j$ is marginal at $\xi := C^f_\Delta(\theta_j, \theta_{-j})$; specifically, $j$ is the next-best grant eligible for one of its categories: the highest-merit element of $A_c(\xi)$ for some $c \in \rho(j)$.*

*Proof.* Were $j$ non-marginal at $\xi$, Proposition 6 would give $C^f_\Delta(\theta'_j, \theta_{-j}) = \xi$, so $j$ would be unfunded under $\theta'_j$ as well, contrary to hypothesis. Hence $j$ is marginal at $\xi$. The first condition of non-marginality holds automatically: $j$ is unfunded at $\xi$, and the algorithm halts only once $F(\xi)$ is empty, so $\underline{b}_j > s(\xi)$. Therefore $j$ fails the second condition, meaning that for some $c \in \rho(j)$ every higher-merit member of $I_c$ is fully funded. Since $\xi_j = 0 < \bar{b}_j$, $j$ is then the highest-merit element of $A_c(\xi)$. □

While the floors-first choice rule is not monotonic in types, we can say exactly where non-monotonicity can occur. By Corollary 1, a grant can lose funding through a monotonic improvement only if it is marginal at the resulting allocation, which here means it is the next-best grant eligible for one of its categories; equivalently, among grants unfunded at $\xi$, only those next-best in some category at $\xi$ can benefit from a monotonic disimprovement. Non-marginal grants are immune in a stronger sense: by Proposition 6, no monotonic disimprovement in their report changes the allocation at all. Moreover, the exposed population is small, and its size does not grow with the number of applicants.

**Corollary 2.** *At any allocation produced by $C^f_\Delta$, at most $|\mathscr{C}|$ unfunded grants can benefit from a monotonic disimprovement, regardless of $|I|$.*

*Proof.* By Corollary 1, any such grant $j$ is the highest-merit element of $A_c(\xi)$ for some $c \in \rho(j)$. Each $A_c(\xi)$ has at most one highest-merit element, so the assignment of an exposed grant to a category witnessing it is injective into $\mathscr{C}$. □

Hence, for a large population of unfunded grants, the proportion which can benefit from a monotonic disimprovement vanishes: the number that can is capped by the number of categories, a quantity fixed by the policymaker rather than by the applicant pool.

To give a clear picture of how the choice rule works, we provide an example. To make matters simpler, we allow full funding only, so instead of using a funding vector $\xi^t$ to indicate funding after step $t$, we will use the set $W^t$ of grants funded after step $t$.

**Example 3.** *Suppose the floor problem $\mathcal{G}$ consists of five grants $1 \succ 2 \succ 3 \succ 4 \succ 5$ given by:*

$$\begin{array}{lll} 1: & b_1 = 1, & \rho(1) = \{\alpha\} \\ 2: & b_2 = 1, & \rho(2) = \{\beta\} \\ 3: & b_3 = 2, & \rho(3) = \{\alpha, \beta\} \\ 4: & b_4 = 1, & \rho(4) = \{\alpha\} \\ 5: & b_5 = 1, & \rho(5) = \{\beta\}, \end{array}$$

*with $\mathscr{C} = \{\alpha, \beta\}$, $\mathcal{B} = 4$, $q_\alpha = 2$, and $q_\beta = 2$.*

***Step 1 of** $C^f_\Delta$: $W^0 = \emptyset$, and $\mathscr{C}^1 = \{\alpha, \beta\}$. Among grants eligible for at least one category in $\mathscr{C}^1$, grant 1 has the highest merit. Therefore, $W^1 = \{1\}$.*

***Step 2 of** $C^f_\Delta$: Now $\mathscr{C}^2 = \{\beta\}$, since adding the next highest-merit grant eligible for $\alpha$, namely grant 3, would cause funds allocated to $\alpha$ to exceed the floor. Among grants eligible for category $\beta$, grant 2 has the highest merit, so $W^2 = \{1, 2\}$.*

***Step 3 of** $C^f_\Delta$: Now $\mathscr{C}^3 = \emptyset$ because for either category, grant 3 is the next highest-merit grant, and its addition would cause either category to have funds in excess of the floor. Hence, in this step, $C^f_\Delta$ chooses the highest-merit grant in $F(\xi^2)$, which happens to be grant 3. Therefore, $W^3 = \{1, 2, 3\}$.*

***Step 4 of** $C^f_\Delta$: Now $F(\xi^3) = \emptyset$, so the algorithm terminates with $C^f_\Delta(\mathcal{G}) = \{1, 2, 3\}$.*

This section constructed a choice rule that meets floors first and then assigns the remaining funds. Its allocations are non-wasteful, respect priorities and satisfy feasible floors. It is not, however, the only rule with these properties, so Section 7 considers a refinement.

## 6.3 Computation

Each step of the floors-first algorithm makes permanent progress of one of three kinds: it meets a category's floor, it fully funds a grant, or it exhausts the budget. Since there are $|\mathscr{C}|$ categories and $|I|$ grants, this bounds the number of steps.

**Proposition 7.** *The floors-first algorithm operates in polynomial time with a total time complexity of* $\mathcal{O}\left((|I| + |\mathscr{C}|) \cdot |I| \cdot |\mathscr{C}|\right)$ *and terminates in at most* $|\mathscr{C}| + |I| + 1$ *steps.*

# 7 Merit-Maximal Choice Rule

The idea of using merit to totally order allocations and selecting the maximum is due to Sönmez and Yenmez (2019a), where it appears implicitly. We adapt it to our setting, where allocations are continuous and the admissible set is defined by Axioms A–C (non-wastefulness, respecting priorities, and satisfying feasible floors), and give the construction explicitly. For a fixed floor problem, any finite set of allocations satisfying a given set of restrictions is totally ordered by merit, as we show below, and therefore has a maximal element. For earmark problems, we have shown that a unique disaggregate allocation satisfies the desiderata (Theorem 1), so the set to be ordered contains just one element. Therefore, this section focuses on floor problems, and hence on choice rules rather than disaggregate choice rules. Since this approach requires a finite set of allocations, we restrict attention to integer floor problems and to choice rules giving integer solutions. First, some definitions are required.

If $\xi$ and $\xi'$ are allocations for a problem $\mathcal{G}$, we say that $\xi$ *dominates* $\xi'$ if $\xi \neq \xi'$ and the highest-merit grant $i^*$ having $\xi_{i^*} \neq \xi'_{i^*}$ satisfies $\xi_{i^*} > \xi'_{i^*}$, and denote this $\xi \triangleright \xi'$.

**Proposition 8.** *Fix a grant-funding problem* $\mathcal{G}$*. Any nonempty finite set* $\Xi$ *of allocations for* $\mathcal{G}$ *is totally ordered by* $\triangleright$ *and contains a maximum* $\xi^*$*.*

*Proof.* If $\xi, \xi' \in \Xi$ and $\xi \neq \xi'$, then, since $I$ is finite and $\succ$ is strict, there is a unique highest-merit grant $i^*$ for which $\xi_{i^*} \neq \xi'_{i^*}$. Exactly one of $\xi_{i^*} > \xi'_{i^*}$ and $\xi_{i^*} < \xi'_{i^*}$ holds. In the first case $\xi \triangleright \xi'$, and in the second $\xi' \triangleright \xi$. Hence any two distinct elements of $\Xi$ are comparable, and exactly one dominates the other; and $\triangleright$ is irreflexive by definition.

It remains to show that $\triangleright$ is transitive. Suppose $\xi \triangleright \xi'$ and $\xi' \triangleright \xi''$, with $h$ the highest-merit grant at which $\xi$ and $\xi'$ differ and $h'$ the highest-merit grant at which $\xi'$ and $\xi''$ differ. Let $j$ be whichever of $h, h'$ has higher merit ($j = h$ if $h = h'$). Every grant of higher merit than $j$ has higher merit than both $h$ and $h'$, so $\xi$, $\xi'$ and $\xi''$ all assign it the same amount. At $j$ itself, $\xi_j \geq \xi'_j \geq \xi''_j$, and at least one of these inequalities is strict: if $j = h$, then $\xi_j > \xi'_j$ and $\xi'_j \geq \xi''_j$ (with equality unless $j = h'$); if $j = h' \neq h$, then $\xi_j = \xi'_j > \xi''_j$. Thus $\xi_j > \xi''_j$, so $\xi \neq \xi''$, $j$ is the highest-merit grant at which $\xi$ and $\xi''$ differ, and $\xi \triangleright \xi''$.

Therefore $\triangleright$ is a strict total order on $\Xi$. Since $\Xi$ is finite and nonempty, it has a maximum (by induction on $|\Xi|$), and the maximum is unique by asymmetry. In other words, there exists a unique $\xi^* \in \Xi$ such that $\xi^* \triangleright \xi$ for all $\xi \in \Xi \setminus \{\xi^*\}$. □

We call a floor problem $\mathcal{G}$ an *integer floor problem* if each item in the tuple is an integer, and we call a choice rule *integer-valued* if it assigns integer values of funding to each grant. Suppose $\tilde{\mathcal{M}}$ is a non-empty collection of integer-valued choice rules defined on a collection of integer floor problems $\Gamma$. Then we say that $C$ is *merit-maximal on* $\tilde{\mathcal{M}}$ if $C$ belongs to $\tilde{\mathcal{M}}$ and for every choice rule $C' \in \tilde{\mathcal{M}}$ having $C' \neq C$, (1) for every $\mathcal{G} \in \Gamma$, $C'(\mathcal{G})$ does not dominate $C(\mathcal{G})$, and (2) there exists an integer floor problem $\mathcal{G} \in \Gamma$ such that $C(\mathcal{G})$ dominates $C'(\mathcal{G})$. Equivalently, a merit-maximal choice rule is *constrained lexicographically efficient*: among the allocations meeting the imposed criteria, it selects the one that is largest in the lexicographic order $\triangleright$ induced by merit.

The motivation is this. If the purpose of the floors-first choice rule is only to find an allocation satisfying Axioms A–C, and $\mathcal{M}$ is the set of choice rules that always do so, then the merit-maximal choice rule on $\mathcal{M}$ is in a sense the best rule in $\mathcal{M}$, while $C^f_\Delta$ is merely *some* rule in $\mathcal{M}$. We show that this reasoning has a drawback: a merit-maximal choice rule on $\mathcal{M}$ cannot be monotonic in cost, merit or categories, nor independent of non-contending grants.

## 7.1 Application to Axioms A–C

Let $\Gamma$ be the set of all integer floor problems, and for each $\mathcal{G} \in \Gamma$, let $\Xi(\mathcal{G})$ be the set of integer allocations $\xi$ satisfying Axioms A–C (non-wastefulness, respecting priorities, and satisfying feasible floors). Then, $\mathcal{M} = \{C : \mathcal{G} \mapsto \xi \in \Xi(\mathcal{G}) \text{ defined on } \Gamma\}$ is the collection of integer-valued choice rules defined on integer floor problems that give allocations satisfying Axioms A–C.

**Proposition 9.** *For any $\Delta$, $C^f_\Delta$ belongs to $\mathcal{M}$.*

*Proof.* Assume there is an integer floor problem $\mathcal{G} \in \Gamma$. Fix a step $t$ of $C^f_\Delta$ and assume that $\xi^t$ is an integer vector; hence, the amount of budget remaining is an integer and so is $d_c(\xi^t)$ for every category $c$. In step $t+1$, compared to $\xi^t$ the tentative allocation $\xi^{t+1}$ either meets a floor $q_c$ exactly, fully funds a grant, or exhausts the budget. In any of these cases, an integer value of funding is assigned. Hence, $\xi^{t+1}$ is an integer vector. Clearly $\xi^0 = \vec{0}$ is an integer vector, so it follows by induction that $C^f_\Delta(\mathcal{G})$ is an integer vector. By Theorem 2, $C^f_\Delta(\mathcal{G})$ also satisfies Axioms A–C, so $C^f_\Delta(\mathcal{G}) \in \Xi(\mathcal{G})$. □

**Proposition 10.** *The choice rule $C^{\text{mm}}_{\mathcal{M}}$ that assigns to each $\mathcal{G} \in \Gamma$ the $\triangleright$-maximum of $\Xi(\mathcal{G})$ is well defined, and it is the unique merit-maximal choice rule on $\mathcal{M}$.*

*Proof.* For each $\mathcal{G} \in \Gamma$, one may define $C^{\text{mm}}_{\mathcal{M}}(\mathcal{G}) = \xi^*$ where $\xi^*$ is the maximum of $\Xi(\mathcal{G})$; this is possible by Proposition 8, since $\Xi(\mathcal{G})$ is finite and, by Proposition 9, nonempty. It is clear that $C^{\text{mm}}_{\mathcal{M}} \in \mathcal{M}$ because $\xi^* \in \Xi(\mathcal{G})$ for each $\mathcal{G} \in \Gamma$. It is merit-maximal on $\mathcal{M}$: for any $C' \in \mathcal{M}$ with $C' \neq C^{\text{mm}}_{\mathcal{M}}$, no $C'(\mathcal{G})$ dominates $C^{\text{mm}}_{\mathcal{M}}(\mathcal{G})$, since

$C'(\mathcal{G}) \in \Xi(\mathcal{G})$ and $C^{\text{mm}}_{\mathcal{M}}(\mathcal{G})$ is the maximum of $\Xi(\mathcal{G})$; and $C'(\mathcal{G}) \neq C^{\text{mm}}_{\mathcal{M}}(\mathcal{G})$ for some $\mathcal{G} \in \Gamma$, where therefore $C^{\text{mm}}_{\mathcal{M}}(\mathcal{G}) \triangleright C'(\mathcal{G})$. It is the only such rule: if $C \in \mathcal{M}$ with $C \neq C^{\text{mm}}_{\mathcal{M}}$, then $C(\mathcal{G}) \neq C^{\text{mm}}_{\mathcal{M}}(\mathcal{G})$ for some $\mathcal{G} \in \Gamma$, and there $C^{\text{mm}}_{\mathcal{M}}(\mathcal{G}) \triangleright C(\mathcal{G})$. Taking $C' = C^{\text{mm}}_{\mathcal{M}}$ in the definition of merit-maximality, $C'(\mathcal{G})$ dominates $C(\mathcal{G})$, so $C$ is not merit-maximal on $\mathcal{M}$. □

To gain a better understanding of $C^{\text{mm}}_{\mathcal{M}}$, we consider an example in which $C^{\text{mm}}_{\mathcal{M}}$ and $C^{f}_{\Delta}$ agree. As usual, we use a set rather than a vector to denote the chosen grants, since $\underline{b}_i = \bar{b}_i$ for all $i$.

**Example 4.** *Consider the grant-funding problem where grants* $1 \succ 2 \succ 3 \succ 4 \succ 5$ *are as follows:*

$$\begin{array}{lll} 1: & b_1 = 13, & \rho(1) = \emptyset \\ 2: & b_2 = 5, & \rho(2) = \{\alpha\} \\ 3: & b_3 = 2, & \rho(3) = \{\alpha\} \\ 4: & b_4 = 2, & \rho(4) = \{\alpha\} \\ 5: & b_5 = 5, & \rho(5) = \{\alpha\}, \end{array}$$

*where* $\mathscr{C} = \{\alpha\}$, $\mathcal{B} = 20$, *and* $q_\alpha = 10$.

We first find $C^{\text{mm}}_{\mathcal{M}}$. Consider the attainable subsets of $I$ (that is, the attainable allocations) that contain grant 1. If any of them satisfies Axioms A–C, then $C^{\text{mm}}_{\mathcal{M}}$ funds grant 1; otherwise, it does not. Since $\mathcal{B} - b_1 = 7$, a subset containing grant 1 is attainable exactly when the other grants in it cost at most 7 in total, which gives ten such subsets. Five of them—$\{1\}$, $\{1,2\}$, $\{1,3\}$, $\{1,4\}$, and $\{1,5\}$—fail Axiom A: each leaves at least 2 of the budget unspent, so adding grant 3 or grant 4, whichever it lacks, yields a larger attainable allocation. Of the other five, $\{1,2,3\}$, $\{1,2,4\}$, and $\{1,3,4\}$ fail Axiom C, while $\{1,3,5\}$ and $\{1,4,5\}$ fail Axiom B. Hence no attainable subset containing grant 1 satisfies Axioms A–C. Exactly one subset that excludes grant 1 satisfies Axioms A–C, namely $\{2,3,4,5\}$, so $C^{\text{mm}}_{\mathcal{M}}(\mathcal{G}) = \{2,3,4,5\}$.

Had Axiom B been omitted from the definition of $\mathcal{M}$, $C^{\text{mm}}_{\mathcal{M}}$ would have selected $\{1,3,5\}$. This would be undesirable, because grant 2 is a strictly better grant than grant 5: it has the same cost and the same categories, but *higher* merit. Axiom B rules out this outcome, and as a result $C^{\text{mm}}_{\mathcal{M}}$ does not fund grant 1.

For $C^{f}_{\Delta}$, we must have $\Delta = (\{\alpha\})$, and grant 2 is chosen in the first step; the remaining steps give $C^{f}_{\Delta}(\mathcal{G}) = C^{\text{mm}}_{\mathcal{M}}(\mathcal{G}) = \{2,3,4,5\}$. The two rules agree in this example, but they do not agree in general.

## 7.2 Limits of Merit-Maximality

Recall, we defined $\mathcal{M} = \{C : \mathcal{G} \mapsto \xi \in \Xi(\mathcal{G}) \text{ defined on } \Gamma\}$ as the collection of integer-valued choice rules defined on integer floor problems that give allocations satisfying

Axioms A–C (non-wasteful, respects priorities, satisfies feasible floors).

**Theorem 3.** *Any choice rule $C \in \mathcal{M}$ which is independent of non-contending grants, or monotonic in cost, merit, or categories, or which is monotonic in types for non-marginal grants is not merit-maximal on $\mathcal{M}$.*

Any such choice rule $C \in \mathcal{M}$ must sometimes lead to outcomes which are dominated by another outcome satisfying Axioms A–C. In practice, this means that a policymaker who wants to reward better applications, or to keep the allocation unaffected by the arrival of applications that are out of contention, must give up efficiency with respect to the total order.

First, we show that $C^{\text{mm}}_{\mathcal{M}}$ fails monotonicity in cost at a non-marginal grant, so it is neither monotonic in cost nor monotonic in types for non-marginal grants.

**Example 5.** *Let $\mathscr{C} = \{\alpha\}$, $\mathcal{B} = 16$ and $q_\alpha = 12$, and let there be three grants $1 \succ 2 \succ 3$ given by*

$$\begin{array}{lll} 1: & [\underline{b}_1, \bar{b}_1] = [1, 10], & \rho(1) = \emptyset \\ 2: & [\underline{b}_2, \bar{b}_2] = [7, 13], & \rho(2) = \{\alpha\} \\ 3: & [\underline{b}_3, \bar{b}_3] = [2, 4], & \rho(3) = \{\alpha\}. \end{array}$$

*So $I_\alpha = \{2, 3\}$, and every component of $\mathcal{G}$ is an integer.*

The merit-maximal allocation is $C^{\text{mm}}_{\mathcal{M}}(\mathcal{G}) = (4, 12, 0)$. However, when grant 3 deviates by increasing its cost bounds to $[2, 6]$, the merit-maximal allocation is $C^{\text{mm}}_{\mathcal{M}}(\mathcal{G}') = (10, 0, 6)$. Therefore, grant 3 benefits using a monotonic disimprovement. Intuitively, once grant 3 can absorb more funding, it can bring $\alpha$ close enough to its floor that grant 2, the next-best grant there, no longer fits without overshooting. The floor then counts as met, more funding is left for grant 1, and the new allocation dominates the old one. (A detailed proof is given in Appendix A.3.)

Because every choice rule we consider depends on merit only through the order $\succ$, and never through the cardinal values $m_i$, this deviation admits two readings: merit falls, or it stays the same while the request grows. Either reading suffices for Theorem 3.

Next, we illustrate that neither $C^f_\Delta$ nor $C^{\text{mm}}_{\mathcal{M}}$ is monotonic in merit or categories:

**Example 6.** *Suppose there are three grants $1 \succ 2 \succ 3$, given by:*

$$\begin{array}{lll} 1: & b_1 = 4, & \rho(1) = \emptyset \\ 2: & b_2 = 2, & \rho(2) = \{\alpha\} \\ 3: & b_3 = 1, & \rho(3) = \{\alpha\}, \end{array}$$

*where $\mathscr{C} = \{\alpha\}$, $\mathcal{B} = 4$, and $q_\alpha = 1$. Further, it must be that $\Delta = (\{\alpha\})$.*

Notice that $C^f_\Delta(\mathcal{G}) = C^{\text{mm}}_{\mathcal{M}}(\mathcal{G}) = \{1\}$. However, if we make the adjustment $\rho(2) = \emptyset$ and call the resulting problem $\mathcal{G}'$, then $C^f_\Delta(\mathcal{G}') = C^{\text{mm}}_{\mathcal{M}}(\mathcal{G}') = \{2, 3\}$, so grant 2 can benefit from failing to report category $\alpha$. If instead we lower the merit of grant 2 so that $1 \succ 3 \succ 2$ and call the resulting problem $\mathcal{G}''$, then $C^f_\Delta(\mathcal{G}'') = C^{\text{mm}}_{\mathcal{M}}(\mathcal{G}'') = \{2, 3\}$, so lowering the merit of grant 2 turns its rejection into an award.

Finally, we show $C^{\text{mm}}_{\mathcal{M}}$ is not independent of non-contending grants.

**Example 7.** *Suppose there are three grants $1 \succ 2 \succ j$, given by:*

$$\begin{array}{lll} 1: & [\underline{b}_1, \bar{b}_1] = [1, 2], & \rho(1) = \emptyset \\ 2: & [\underline{b}_2, \bar{b}_2] = [2, 3], & \rho(2) = \{\alpha\} \\ j: & [\underline{b}_j, \bar{b}_j] = [1, 1], & \rho(j) = \{\alpha\}, \end{array}$$

*where $\mathscr{C} = \{\alpha\}$, $\mathcal{B} = 3$, and $q_\alpha = 2$.*

In the problem with only $I = \{1, 2\}$, the allocation is $C^{\text{mm}}_{\mathcal{M}}(\mathcal{G}) = (1, 2)$, whereas in the augmented problem, $\mathcal{G}^{+j}$, the allocation is $C^{\text{mm}}_{\mathcal{M}}(\mathcal{G}^{+j}) = (2, 0, 1)$. Hence, the rule fails independence of non-contending grants. Grant $j$ is non-marginal at $C^{\text{mm}}_{\mathcal{M}}(\mathcal{G}) = (1, 2)$: its minimum cost exceeds the leftover funds, which are zero, and $\alpha$, the only category it is eligible for, has a higher-merit eligible grant, grant 2, that is not fully funded. It is therefore non-contending there, since grants 1 and 2 are the only funded grants and both rank above $j$.

**Corollary 3.** *For any $\Delta$, $C^f_\Delta$ is not merit-maximal on $\mathcal{M}$.*

*Proof.* By Proposition 9, $C^f_\Delta \in \mathcal{M}$, and by Theorem 2, $C^f_\Delta$ is monotonic in cost. Theorem 3 then implies that $C^f_\Delta$ is not merit-maximal on $\mathcal{M}$. □

The merit order determines a unique merit-maximal choice rule on $\mathcal{M}$, namely $C^{\text{mm}}_{\mathcal{M}}$ (Proposition 10); however, $C^{\text{mm}}_{\mathcal{M}}$ is neither monotonic in cost, merit, or categories nor independent of non-contending grants. By Theorem 3, no choice rule in $\mathcal{M}$ that is monotonic in cost or independent of non-contending grants is merit-maximal on $\mathcal{M}$; merit-maximality must therefore be relinquished to obtain either property. The floors-first choice rule is not merit-maximal (Corollary 3), but it satisfies both properties (Theorem 2). On each incentive and robustness property we consider, $C^f_\Delta$ does at least as well as $C^{\text{mm}}_{\mathcal{M}}$, and on monotonicity in cost, monotonicity in types for non-marginal grants, and independence of non-contending grants it does strictly better; what $C^{\text{mm}}_{\mathcal{M}}$ offers in return is merit-maximality alone.

## 7.3 Computation

The merit-maximal choice rule selects the maximum of a set of allocations whose size grows exponentially in $|I|$, and we know of no implementation of it that avoids

enumerating that set. The floors-first choice rule, by contrast, is implementable in polynomial time. This is a further respect, alongside monotonicity in cost and independence of non-contending grants, in which $C^f_\Delta$ may be preferred to $C^{\text{mm}}_{\mathcal{M}}$ in practice.

# 8 Conclusion

Priority design extends from indivisible assignment problems to settings where divisible funds are allocated to recipients that have minimum costs and may be eligible for many funding categories. When funding is earmarked, we characterize the solution by requiring that eligible categories contribute equally to a grant wherever they can. Floors, on the other hand, cannot always be met in this setting, so we work with a relaxed notion, feasible floors. Conceptually, a floor problem does not designate which funds count towards which category, and the relaxed notion of floors admits a wider set of acceptable allocations; together these leave enough degrees of freedom that our axioms do not single out a rule.

We therefore turn to a refinement. The merit-maximal choice rule, which is constrained lexicographically efficient for integer floor problems, is unique, but relative to the floors-first choice rule it has inferior incentive properties and is sensitive to the arrival of grants that were never in contention for funding. Since efficiency comes at this cost, we recommend the floors-first choice rule: it is monotonic in cost, it is monotonic in types except at marginal grants, whose number is bounded by the number of categories, and it is independent of non-contending grants. Its appeal is also conceptual, since it settles every funding decision at one of the two margins identified in Section 4.5: the budget margin, funds left idle, and the merit frontier, the best grant not yet fully funded within each category and overall. Like the serial-SR choice rule, it can be implemented in polynomial time.

Which scheme is appropriate depends on the institution. Where funding carries strict designations—a donor who restricts how a gift may be spent, or a line item appropriated for a stated purpose—earmarks are the right instrument, and the cost of a grant must be apportioned among the categories that support it. Where instead a policymaker wants a balanced portfolio, and the setting permits a funded project to count in full towards every category it is eligible for, floors are the appropriate constraint.

Our model takes the merit ranking as given rather than modeling the two-tiered review that produces it, and it treats reported cost bounds as observable and the allocation as one-shot. Relaxing these assumptions, and incorporating other institutional features, suggests several directions for future research. One is "soft reserves," in which funds are set aside for a category but may be used for any grant once every

grant in that category has been funded. Another is dynamic allocation, in which a rule must weigh the opportunity cost of committing funds now against waiting for later applications. Finally, two questions about floor problems remain open: whether Axioms A, B, and C* can always be satisfied simultaneously (Appendix A.4), and whether the choice rules for floor problems admit an axiomatic characterization. Minimum costs considerably complicate the analysis, but they supply the realism that budgeting in practice demands. More broadly, our results show that the tools of priority design extend to the allocation of divisible funds among recipients with minimum costs, and we hope they provide a foundation for further work on grant funding.

## Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude (Anthropic) in order to edit the text and notation for clarity and consistency, to draft and restructure parts of the exposition and of the proofs in Appendix A, to check proofs and examples for errors, and to write the Python code in the replication materials, which checks the allocations in the proof of Proposition 2 and carries out the search reported in Appendix A.4. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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# A Proofs

## A.1 Earmarks

Throughout this subsection the grants are listed in merit order, $i_1 \succ \cdots \succ i_n$, as in the serial-SR algorithm; $x^t$ is the matrix after step $t$, and $g = i_t$ is the grant considered

at step $t$, for whichever step is under discussion. Each row is set at its own step and never changed afterwards, so $x_g = x_g^t$. For a category $c$, write

$$u_c = q_c - x_c^{t-1}$$

for the reserve of $c$ still unassigned when $g$ is considered, so that $r_g(x^{t-1}) = \sum_{c\in\rho(g)} u_c$ is the total reserve available to $g$ at its step. Since column sums never fall, $q_c - x_c \le u_c$ at the end of the run.

**Lemma 1.** *Fix a step $t$.*

*(a) If $r_g(x^{t-1}) < \underline{b}_g$, the row of $g$ is zero. Otherwise $x_g = \min\{\bar{b}_g,\, r_g(x^{t-1})\}$, which lies in $[\underline{b}_g, \bar{b}_g]$, and (2)–(3) have exactly one solution, $x_{g,c}^t = \min\{u_c, \lambda\}$ for $c \in \rho(g)$, where $\lambda$ is the unique number in $\left(0, \max_{c\in\rho(g)} u_c\right]$ with $\sum_{c\in\rho(g)} \min\{u_c, \lambda\} = x_g$. In particular, $\lambda = \max_{c\in\rho(g)} x_{g,c}$.*

*(b) $x^t$ is attainable; in particular, the outcome $x$ of the algorithm is an attainable disaggregate allocation.*

*(c) If $\underline{b}_g \le x_g < \bar{b}_g$, every category in $\rho(g)$ is exhausted at the end of step $t$: $x_c^t = q_c$ for all $c \in \rho(g)$.*

*(d) For every category $c$, $x_c^{t-1} = \sum_{i\in\hat{I}_{g,c}} x_{i,c}$; that is, $u_c$ is what the grants of higher merit than $g$ that are eligible for $c$ leave of $q_c$.*

*Proof of Lemma 1.* (a) The first statement is the algorithm's rule. Suppose $r_g(x^{t-1}) \ge \underline{b}_g$. Then (2) sets $x_g = \min\{\bar{b}_g, r_g(x^{t-1})\}$, and $0 < \underline{b}_g \le x_g \le r_g(x^{t-1})$ because $\bar{b}_g \ge \underline{b}_g$. Write $\bar{u} = \max_{c\in\rho(g)} u_c$, and let $\phi(s) = \sum_{c\in\rho(g)} \min\{u_c, s\}$. Then $\phi$ is continuous, $\phi(0) = 0$, $\phi$ is strictly increasing on $[0, \bar{u}]$ and constant on $[\bar{u}, \infty)$, and $\phi(\bar{u}) = r_g(x^{t-1})$. Hence there is exactly one $\lambda \in (0, \bar{u}]$ with $\phi(\lambda) = x_g$. The row $x_{g,c}^t = \min\{u_c, \lambda\}$ sums to $x_g$, which is (2), and its maximum is $\min\{\bar{u}, \lambda\} = \lambda$, which gives (3). Conversely, if a row solves (3) and $\mu$ is its maximum, then $x_{g,c}^t = \min\{u_c, \mu\}$ for every $c$, and taking the maximum over $c$ gives $\mu = \min\{\bar{u}, \mu\} \le \bar{u}$; (2) then reads $\phi(\mu) = x_g$, so $\mu = \lambda$.

(b) Suppose $x^{t-1}$ is attainable, as $x^0 = 0$ is. By (a), the row of $g$ is either 0 or sums to a value in $[\underline{b}_g, \bar{b}_g]$, and $x_{g,c}^t \le u_c$, so no column sum exceeds its reserve. The other rows are unchanged.

(c) If $x_g < \bar{b}_g$, then $x_g = r_g(x^{t-1}) = \sum_{c\in\rho(g)} u_c$ by (a). Since $x_{g,c}^t \le u_c$ for each $c$ and the $x_{g,c}^t$ sum to $x_g$, every inequality binds: $x_{g,c}^t = u_c$, that is, $x_c^t = q_c$.

(d) The grants considered before step $t$ are exactly those of higher merit than $g$, and their rows are final; the grants considered later still have zero rows. By construction a row is zero outside the grant's own categories, so the grants contributing to $x_c^{t-1}$ are those in $\hat{I}_{g,c}$. □

We use Lemma 1 in the proofs of Theorem 1, Proposition 3, Proposition 4 and Proposition 5.

*Proof of Theorem 1.* By Lemma 1(a) and (b), $C^{\mathrm{SR}}$ is a well-defined disaggregate choice rule. It remains to show that its outcome $x$ satisfies the four axioms, and that no other disaggregate allocation does.

**Claim 1.1.** *$x$ satisfies Axioms A, B, D and E.*

*Proof. Axiom D.* By construction, $x_{i,c} > 0$ only if $c \in \rho(i)$.

*Axiom A.* Since $x$ is eligibility-compliant, it suffices to check (1) with $r_g(x)$ for each grant $g$. If $x_g = 0$, then $r_g(x^{t-1}) < \underline{b}_g$ by Lemma 1(a), and $r_g(x) \leq r_g(x^{t-1})$. If $\underline{b}_g \leq x_g < \bar{b}_g$, then $r_g(x) = 0$ by Lemma 1(c).

*Axiom B, first clause.* Let $x_g = 0$ and take any $J \subseteq I$. Since $g \,\pi_c\, j$ holds only if $g$ and $j$ are both eligible for $c$ and $g \succ j$, a unit counted by $\sum_{j\in J} \sum_{c\,:\, g\,\pi_c\, j} x_{j,c}$ belongs to a category $c \in \rho(g)$ and to a grant $j$ of lower merit than $g$, so it was assigned after step $t$. Since $g$ received nothing, for each $c \in \rho(g)$ these units and the reserve $q_c - x_c$ left at the end together amount to at most $q_c - x_c^t = u_c$. Summing over $c \in \rho(g)$,

$$\sum_{j\in J} \sum_{c\,:\, g\,\pi_c\, j} x_{j,c} + \sum_{c\in\rho(g)} \big(q_c - x_c\big) \;\leq\; r_g(x^{t-1}) \;<\; \underline{b}_g,$$

so no $J$ violates the clause.

*Axiom B, second clause.* Let $\underline{b}_g \leq x_g < \bar{b}_g$. By Lemma 1(c) every category in $\rho(g)$ is exhausted after step $t$, so no grant considered later, that is, no grant of lower merit, receives anything from $\rho(g)$.

*Axiom E.* Let $c \in \rho(g)$ with $x_{g,c} < \max_{c'\in\rho(g)} x_{g,c'}$. Then $g$ is funded, and by Lemma 1(a) $x_{g,c} = \min\{u_c, \lambda\} < \lambda$, so $x_{g,c} = u_c$. By Lemma 1(d), this is $x_{g,c} + \sum_{i\in\hat{I}_{g,c}} x_{i,c} = q_c$. □

**Claim 1.2.** *Any disaggregate allocation $x'$ satisfying Axioms A, B, D and E coincides with $x$.*

*Proof.* We show by induction on $t$ that the row of $g = i_t$ is the same in $x'$ as in $x$. Suppose this holds for $i_1, \ldots, i_{t-1}$. Then the grants of higher merit than $g$ hold the same amounts in $x'$ as in $x$, namely $x_c^{t-1}$ of each category $c$, and by Lemma 1(d) this is $\sum_{i\in\hat{I}_{g,c}} x'_{i,c}$. Hence, for each $c \in \rho(g)$, the reserve $u_c$ is split in $x'$ among $g$, the grants of lower merit, and what is left unassigned:

$$u_c = x'_{g,c} + \sum_{j\,:\, g\succ j} x'_{j,c} + \big(q_c - x'_c\big), \tag{4}$$

with every term nonnegative because $x'$ is attainable. Summing over $c \in \rho(g)$ and using Axiom D,

$$r_g(x^{t-1}) = x'_g + L + r_g(x'), \qquad L = \sum_{j\,:\,g \succ j} \sum_{c \in \rho(g)} x'_{j,c}.$$

Since $x'$ is attainable, $x'_g \le \bar{b}_g$, and the display gives $x'_g \le r_g(x^{t-1})$.

*The total is the same.* If $r_g(x^{t-1}) < \underline{b}_g$, then $x'_g \le r_g(x^{t-1}) < \underline{b}_g$ forces $x'_g = 0 = x_g$. Otherwise $x_g = \min\{\bar{b}_g, r_g(x^{t-1})\} \ge x'_g$ by Lemma 1(a); suppose $x'_g < x_g$, so that $L + r_g(x') = r_g(x^{t-1}) - x'_g > 0$.

- If $x'_g = 0$ and $L = 0$, then $r_g(x') = r_g(x^{t-1}) \ge \underline{b}_g$, and $x'$ fails Axiom A.
- If $x'_g = 0$ and $L > 0$, some grant $j$ of lower merit holds funds from a category $c \in \rho(g)$; by Axiom D, $c \in \rho(j)$, so $g\,\pi_c\,j$. Take $J = \{j \in I : g \succ j\}$. For $j \in J$, $g\,\pi_c\,j$ holds exactly when $c \in \rho(g) \cap \rho(j)$, so by Axiom D the sum in the first clause of Axiom B equals $L + r_g(x') = r_g(x^{t-1}) \ge \underline{b}_g$, and $x'$ fails that clause.
- If $x'_g > 0$, then $\underline{b}_g \le x'_g < x_g \le \bar{b}_g$. If $r_g(x') > 0$, $x'$ fails Axiom A; if $L > 0$, some grant of lower merit draws on $\rho(g)$, and $x'$ fails the second clause of Axiom B.

Hence $x'_g = x_g$.

*The split is the same.* If $x_g = 0$, both rows are zero. Otherwise let $\mu = \max_{c \in \rho(g)} x'_{g,c}$. For $c \in \rho(g)$, (4) gives $x'_{g,c} \le u_c$, and if $x'_{g,c} < \mu$, Axiom E for $x'$ gives $x'_{g,c} + \sum_{i \in \hat{I}_{g,c}} x'_{i,c} = q_c$, that is, $x'_{g,c} = u_c$. So $x'_{g,c} = \min\{u_c, \mu\}$ for every $c \in \rho(g)$: the row of $g$ in $x'$ solves (3), and by the previous step it sums to $x_g$, so it solves (2). By Lemma 1(a) the solution is unique, and by Axiom D both rows vanish outside $\rho(g)$. Hence the rows coincide. □

Claims 1.1 and 1.2 establish that $C^{\mathrm{SR}}$ yields the unique disaggregate allocation satisfying Axioms A, B, D and E. □

*Proof of Proposition 2.* In each case $o$ denotes the open category and $\alpha$ a reserved one, and grants are listed in merit order.[4]

**Axiom A.** Let $I = \{1\}$ and $\mathscr{C} = \{o\}$ with $q_o = 2 = \mathcal{B}$, where $\rho(1) = \{o\}$ and $[\underline{b}_1, \bar{b}_1] = [1, 2]$. Take $x_{1,o} = 1$. Eligibility-compliance is immediate, and with a single eligible category Axiom E is vacuous. The one grant has nothing ranked below it, so

[4] A script checking the four axioms against these four allocations directly, as they are stated in Sections 4.3 and 4.6, is included in the supplementary material and deposited at Mendeley Data (Jourden, 2026); a development copy is at https://github.com/joejourden/priority-design-grant-funding. Like the search code of Appendix A.4, it was written with the assistance of Claude (Anthropic) and reviewed by the author.

both clauses of Axiom B hold vacuously. But $x'_{1,o} = 2$ is attainable and eligibility-compliant with $x' \geq x$, so Axiom A fails.

**Axiom B.** Let $I = \{1, 2\}$ with $1 \succ 2$ and $\mathscr{C} = \{o\}$ with $q_o = 2 = \mathcal{B}$, where $\rho(1) = \rho(2) = \{o\}$ and $[\underline{b}_i, \bar{b}_i] = [2, 2]$ for both. Take $x_{2,o} = 2$ and $x_{1,o} = 0$. The reserve is exhausted, so no grant can absorb more and Axiom A holds; compliance is immediate and Axiom E is again vacuous. But grant 1 is unfunded while grant 2, which it outranks in $\pi_o$, holds $2 \geq \underline{b}_1$, so the first clause of Axiom B fails.

**Axiom D.** Let $I = \{1\}$ and $\mathscr{C} = \{o, \alpha\}$ with $q_o = q_\alpha = 1$ and $\mathcal{B} = 2$, where $\rho(1) = \{o\}$ and $[\underline{b}_1, \bar{b}_1] = [2, 2]$. Take $x_{1,o} = x_{1,\alpha} = 1$. Both reserves are exhausted and grant 1 is fully funded, so Axioms A and B hold, and Axiom E quantifies over $\rho(1) = \{o\}$ alone and is vacuous. But $\alpha \notin \rho(1)$ while $x_{1,\alpha} > 0$, so Axiom D fails.

**Axiom E.** Let $I = \{1\}$ and $\mathscr{C} = \{o, \alpha\}$ with $q_o = q_\alpha = 2$ and $\mathcal{B} = 4$, where $\rho(1) = \{o, \alpha\}$ and $[\underline{b}_1, \bar{b}_1] = [2, 2]$. Take $x_{1,o} = 2$ and $x_{1,\alpha} = 0$. Grant 1 is fully funded, so nothing can absorb the reserve standing in $\alpha$ and Axiom A holds; there is no unfunded grant and none funded below its maximum cost, so Axiom B holds, and compliance is immediate. But $x_{1,\alpha} = 0 < 2 = x_{1,o}$ while $\alpha$ retains its entire reserve, so Axiom E fails. □

*Proof of Proposition 3.* Let $\mathcal{G}'$ equal $\mathcal{G}$ except that grant $i$ reports a weakly worse type, with categories $\rho'(i) \subseteq \rho(i)$ and minimum cost $\underline{b}'_i \geq \underline{b}_i$, and let $x = C^{\mathrm{SR}}(\mathcal{G})$ and $x' = C^{\mathrm{SR}}(\mathcal{G}')$. Let $t$ and $t'$ be the steps at which $i$ is considered in $\mathcal{G}$ and $\mathcal{G}'$; since $i$'s merit can only fall, $t \leq t'$. The first $t-1$ steps consider the same grants with the same data in both problems, so $x^{t-1} = x'^{t-1} \leq x'^{t'-1}$, the inequality because funding only accumulates. Suppose $x'_i > 0$. By Lemma 1(a) in $\mathcal{G}'$,

$$\sum_{c \in \rho(i)} \left(q_c - x_c^{t-1}\right) \;\geq\; \sum_{c \in \rho'(i)} \left(q_c - x_c'^{t'-1}\right) \;\geq\; \underline{b}'_i \;\geq\; \underline{b}_i,$$

so $x_i > 0$ by Lemma 1(a) in $\mathcal{G}$. □

*Proof of Proposition 4.* Let $s$ be the position of $j$ in $\mathcal{G}^{+j}$, so that $j$ ranks below $i_1, \ldots, i_{s-1}$ and above $i_s, \ldots, i_n$. The first $s-1$ steps are the same in both problems, so they produce the same rows and the same matrix $x^{s-1}$. By hypothesis every funded grant outranks $j$, so $i_s, \ldots, i_n$ receive nothing in $\mathcal{G}$; hence $x = x^{s-1}$, and by Lemma 1(a) each $i_k$ with $k \geq s$ has $r_{i_k}(x) < \underline{b}_{i_k}$. In $\mathcal{G}^{+j}$ these grants come after $j$ and face reserves no larger than $q_c - x_c$, so they again receive nothing. Hence $C^{\mathrm{SR}}(\mathcal{G}^{+j})_i = x_i$ for every $i \in I$. If in addition $\underline{b}_j > r_j(x)$, then $j$ faces exactly the reserves $q_c - x_c$ at its step and, by Lemma 1(a), receives nothing, so $C^{\mathrm{SR}}(\mathcal{G}^{+j}) = (x, 0)$. □

*Proof of Proposition 5.* We take the merit order as given, as review delivers it; sorting the grants into that order would add $O(|I| \log |I|)$. Each step settles one row, so the

algorithm halts after $|I|$ steps. At step $t$, write $k = |\rho(g)| \le |\mathscr{C}|$. Computing the residuals $u_c$, their total $r_g(x^{t-1})$, and the comparison with $\underline{b}_g$ costs $O(k)$. If $g$ is funded, Lemma 1(a) reduces the step to finding the level $\lambda$ with $\phi(\lambda) = x_g$, where $x_g = \min\{\bar{b}_g, r_g(x^{t-1})\}$ and $\phi$ is as in the proof of Lemma 1. Sort the residuals as $u_{(1)} \le \cdots \le u_{(k)}$, set $u_{(0)} = 0$ and $P_j = \sum_{l \le j} u_{(l)}$. On $[u_{(j-1)}, u_{(j)}]$ the function $\phi$ is linear, $\phi(\lambda) = P_{j-1} + (k-j+1)\lambda$, and $\phi(u_{(j)}) = P_j + (k-j)u_{(j)}$. So if $j^*$ is the first $j$ with $\phi(u_{(j)}) \ge x_g$, then $\lambda = (x_g - P_{j^*-1})/(k-j^*+1)$. Sorting costs $O(k \log(k+1))$; the prefix sums, the scan for $j^*$, setting $x^t_{g,c} = \min\{u_c, \lambda\}$ and updating the column sums cost $O(k)$ each. Summing over the $|I|$ steps gives $O(|I| \cdot |\mathscr{C}| \cdot \log(|\mathscr{C}|+1))$. □

## A.2 Floors

Two lemmas will aid in the proof of Theorem 2. The first shows that expanding an allocation which satisfies Axiom C cannot lead to an allocation failing to satisfy Axiom C. The second shows that procedures of the kind $C^f_\Delta$ runs in its floor phase terminate at attainable allocations that satisfy feasible floors. Its proof charges each increment of funding to a single category and tracks these charges cumulatively, which yields both the budget bound and the slack that keeps each selected grant in $F(\xi^{t-1})$.

**Lemma 2.** *If $\xi^A$ and $\xi^B$ are two attainable allocations, where $\xi^A \le \xi^B$ and $\xi^A$ satisfies Axiom C, then so does $\xi^B$.*

*Proof of Lemma 2.* Fix a category $c$. If $d_c(\xi^B) \le 0$, Axiom C holds at $c$ for $\xi^B$, so suppose $d_c(\xi^B) > 0$. Since $\xi^A \le \xi^B$, $d_c(\xi^A) \ge d_c(\xi^B) > 0$. Attainability gives $\xi^B_i \le \bar{b}_i$ for every $i$, so a grant fully funded at $\xi^A$ is fully funded at $\xi^B$, and $A_c(\xi^B) \subseteq A_c(\xi^A)$. If $A_c(\xi^B) = \emptyset$, Axiom C holds vacuously at $c$. Otherwise $A_c(\xi^A) \ne \emptyset$, and because $\xi^A$ satisfies Axiom C at $c$, its highest-merit element $i^*$ has $\xi^A_{i^*} = 0$ and $\underline{b}_{i^*} > d_c(\xi^A)$. If $\xi^B_{i^*} > 0$, then $\xi^B_{i^*} \ge \underline{b}_{i^*}$ by attainability, and since $i^* \in I_c$ and $\xi^B \ge \xi^A$,

$$d_c(\xi^B) \;\le\; d_c(\xi^A) - \xi^B_{i^*} \;\le\; d_c(\xi^A) - \underline{b}_{i^*} \;<\; 0,$$

a contradiction. Hence $\xi^B_{i^*} = 0 < \bar{b}_{i^*}$, so $i^* \in A_c(\xi^B)$, and as the highest-merit element of the larger set $A_c(\xi^A)$ it is the highest-merit element of $A_c(\xi^B)$. Finally $\underline{b}_{i^*} > d_c(\xi^A) \ge d_c(\xi^B)$, so Axiom C holds at $c$ for $\xi^B$. □

**Lemma 3.** *Let $\xi^0 = 0$. At each step $t \ge 1$, if there are categories $c$ such that $\xi^{t-1}$ does not satisfy Axiom C for $c$, let $\mathscr{C}^t$ be a nonempty subset of them; otherwise, $\mathscr{C}^t = \emptyset$. If $\mathscr{C}^t$ is empty, let $\xi^t = \xi^{t-1}$, and the procedure ends. Otherwise, choose a grant $g$ and a category $e \in \mathscr{C}^t \cap \rho(g)$ such that*

*(1) $\xi^{t-1}_g < \bar{b}_g$,*

*(2)* $d_e(\xi^{t-1}) > 0$, *and*

*(3) if* $\xi_g^{t-1} = 0$ *then also* $\underline{b}_g \le d_e(\xi^{t-1})$,

*and let*

$$\xi_g^t = \xi_g^{t-1} + \min\left\{\bar{b}_g - \xi_g^{t-1},\ \max_{c \in \mathscr{C}^t \cap \rho(g)} d_c(\xi^{t-1})\right\},$$

*with* $\xi_i^t = \xi_i^{t-1}$ *for all* $i \neq g$. *We write* $g$ *and* $e$ *for the grant and category chosen at whichever step is under discussion. (Such a pair exists: take any* $c \in \mathscr{C}^t$ *and let* $g$ *be the highest-merit element of* $A_c(\xi^{t-1})$, *with* $e = c$. *Since* $\xi^{t-1}$ *fails Axiom C at* $c$, $d_c(\xi^{t-1}) > 0$ *and* $A_c(\xi^{t-1}) \neq \emptyset$, *which gives (1) and (2); and if* $\xi_g^{t-1} = 0$, *failure of Axiom C at* $c$ *means* $\underline{b}_g \le d_c(\xi^{t-1})$, *which is (3).) Any procedure of this type ends at a step* $T \le |I| + |\mathscr{C}| + 1$. *Moreover, for every* $t \le T$, $\xi^t$ *is attainable and*

$$\sum_{i \in I} \xi_i^t + d_c(\xi^t) \ \le\ \sum_{c' \in \mathscr{C}} q_{c'} \ \le\ \mathcal{B} \qquad \text{for every } c \in \mathscr{C}, \tag{5}$$

*and* $\xi^T$ *satisfies Axiom C.*

*Proof of Lemma 3.* Write $S_c(\xi) = \sum_{i \in I_c} \xi_i$, so that $d_c(\xi) = q_c - S_c(\xi)$. For each non-terminal step $t$, let $\gamma^t = \xi_g^t - \xi_g^{t-1}$ be the increment, and fix a category

$$\hat{e} \in \operatorname*{argmax}_{c \in \mathscr{C}^t \cap \rho(g)} d_c(\xi^{t-1}), \qquad \text{so that} \qquad \gamma^t = \min\left\{\bar{b}_g - \xi_g^{t-1},\ d_{\hat{e}}(\xi^{t-1})\right\}.$$

Note $g \in I_{\hat{e}}$, since $\hat{e} \in \rho(g)$.

**Step 1: one-step bounds.** Fix a non-terminal step $t$.

(a) $\gamma^t \le d_{\hat{e}}(\xi^{t-1})$ and $\xi_g^t \le \bar{b}_g$, directly from the definition.

(b) $\gamma^t > 0$: the first term is positive by (1), and $d_{\hat{e}}(\xi^{t-1}) \ge d_e(\xi^{t-1}) > 0$ by maximality and (2).

(c) If $\xi_g^{t-1} = 0$, then $\gamma^t \ge \underline{b}_g$: the first term is $\bar{b}_g \ge \underline{b}_g$, and condition (3) gives $\underline{b}_g \le d_e(\xi^{t-1}) \le d_{\hat{e}}(\xi^{t-1})$.

Funding is nondecreasing along the run. By (c), a grant's first increment carries it to at least its minimum cost, and by (a) no increment carries it past its maximum cost. Hence for every $t$ and every $i$, either $\xi_i^t = 0$ or $\underline{b}_i \le \xi_i^t \le \bar{b}_i$.

**Step 2: cumulative charges.** Charge the increment $\gamma^t$ to the category $\hat{e}$ chosen at step $t$, and let

$$\kappa_c^t = \sum_{s \le t\,:\, c \text{ is charged at step } s} \gamma^s$$

be the total charged to $c$ through step $t$, with $\kappa_c^0 = 0$. A grant can be selected at several steps; $\kappa_c^t$ counts every increment charged to $c$, whichever grant received it and however many times that grant was funded. Three facts follow.

(i) $\sum_{i\in I}\xi_i^t = \sum_{c\in\mathscr{C}}\kappa_c^t$. Each non-terminal step changes one coordinate, by $\gamma^t$, and charges $\gamma^t$ to exactly one category; and $\xi^0 = 0$.

(ii) $\kappa_c^t \le S_c(\xi^t)$. Every increment charged to $c$ was paid to a grant in $I_c$, and $S_c(\xi^t)$ is the sum of all increments paid to grants in $I_c$ through step $t$, all of them nonnegative.

(iii) $\kappa_c^t \le q_c$. By induction on $t$, the case $t = 0$ being trivial. If $\hat{e} \ne c$, then $\kappa_c^t = \kappa_c^{t-1} \le q_c$. If $\hat{e} = c$, then by (ii) at $t-1$ and Step 1(a),

$$\kappa_c^t = \kappa_c^{t-1} + \gamma^t \le S_c(\xi^{t-1}) + d_c(\xi^{t-1}) = q_c.$$

Combining, for any $t$ and any category $c$,

$$\sum_{i\in I}\xi_i^t = \kappa_c^t + \sum_{c'\ne c}\kappa_{c'}^t \le S_c(\xi^t) + \sum_{c'\ne c} q_{c'} = \sum_{c'\in\mathscr{C}} q_{c'} - d_c(\xi^t),$$

which is (5); the second inequality there is the standing assumption $\sum_c q_c \le \mathcal{B}$ on floor problems. Directly from (i) and (iii), $\sum_i \xi_i^t = \sum_c \kappa_c^t \le \sum_c q_c \le \mathcal{B}$. Together with Step 1, $\xi^t$ is attainable.

**Step 3: termination and Axiom C.** At each non-terminal step one of two things happens. If $\gamma^t = \bar{b}_g - \xi_g^{t-1}$, grant $g$ becomes fully funded and, by (1), is never selected again. Otherwise $\gamma^t = d_{\hat{e}}(\xi^{t-1})$, so $d_{\hat{e}}(\xi^t) = 0$; as funding is nondecreasing, $d_{\hat{e}} \le 0$ at all later steps, so $\hat{e}$ satisfies Axiom C from then on and never again belongs to $\mathscr{C}^s$, and in particular is never again charged. Hence there are at most $|I| + |\mathscr{C}|$ non-terminal steps, and $T \le |I| + |\mathscr{C}| + 1$. The procedure ends only when $\mathscr{C}^T = \emptyset$, that is, when $\xi^{T-1} = \xi^T$ satisfies Axiom C for every category. □

*Proof of Theorem 2.* To show the function is a choice rule, it suffices to show that the procedure terminates in finite time and it has attainable outcomes.

**Finite termination, attainable outcomes, and Axiom C (satisfying feasible floors).** Call the steps of $C_\Delta^f$ with $\mathscr{C}^t \ne \emptyset$ the *floor phase* and the remaining non-terminal steps the *open phase*.

*The floor phase is a run of the procedure in Lemma 3, and it never stalls on the budget.* Suppose $\xi^{t-1}$ was reached by floor-phase steps that are steps of that procedure, and $\mathscr{C}^t \ne \emptyset$. The grant $g$ chosen by $C_\Delta^f$ is the highest-merit element of $\bigcup_{c\in\mathscr{C}^t} A_c(\xi^{t-1})$, so $g \in A_c(\xi^{t-1})$ for some $c \in \mathscr{C}^t$; fix such a $c$ and set $e = c$. Then $g$ is also the highest-merit element of $A_e(\xi^{t-1})$, so conditions (1)–(3) of Lemma 3 hold by the existence argument given there, and the increment prescribed by $C_\Delta^f$, $\min\{\bar{b}_g - \xi_g^{t-1}, M_t\}$ with $M_t = \max_{c\in\mathscr{C}^t\cap\rho(g)} d_c(\xi^{t-1})$, is exactly the one in Lemma 3.

It remains to check that $C^f_\Delta$ does not halt here, i.e. that $F(\xi^{t-1}) \neq \emptyset$; we show $g \in F(\xi^{t-1})$. Apply (5) at $t-1$ to the category $e$, which has $d_e(\xi^{t-1}) > 0$ by (2):

$$d_e(\xi^{t-1}) \leq s(\xi^{t-1}).$$

If $\xi^{t-1}_g = 0$, condition (3) gives $\underline{b}_g \leq d_e(\xi^{t-1}) \leq s(\xi^{t-1})$. If $\xi^{t-1}_g > 0$, attainability of $\xi^{t-1}$ and (1) give $\underline{b}_g \leq \xi^{t-1}_g < \bar{b}_g$, and $s(\xi^{t-1}) \geq d_e(\xi^{t-1}) > 0$. In either case $g \in F(\xi^{t-1})$. The selected increment itself also fits the remaining budget, since $\xi^t$ is attainable by Lemma 3. By induction, the floor phase of $C^f_\Delta$ coincides with a run of the procedure in Lemma 3, never finds $F(\xi^{t-1})$ empty, and ends after some step $l \leq |I| + |\mathscr{C}|$ at an attainable $\xi^l$ that satisfies Axiom C.

*The open phase.* Since $\xi^{t-1} \geq \xi^l$ for every $t > l$, Lemma 2 gives $\mathscr{C}^t = \emptyset$ at every later step, so once the open phase begins it never returns to the floor phase, and Axiom C holds at every later allocation. At an open-phase step, $g \in F(\xi^{t-1})$ and $M_t = s(\xi^{t-1})$, so the total after the step is at most $\mathcal{B}$ and $\xi^t_g \leq \bar{b}_g$. If $\xi^{t-1}_g = 0$, membership in $F(\xi^{t-1})$ gives $\underline{b}_g \leq M_t$, and $\underline{b}_g \leq \bar{b}_g$, so $\xi^t_g \geq \underline{b}_g$. Hence attainability is preserved. Each open-phase step either fully funds $g$ or brings the total to $\mathcal{B}$; in the latter case $s(\xi^t) = 0$, so $F(\xi^t)$ is empty because $\underline{b}_i > 0$ for every $i$, and $C^f_\Delta$ halts at the next step. So the open phase has finitely many steps, and $C^f_\Delta$ terminates at an attainable allocation satisfying Axiom C.

**Axiom A (non-wastefulness).** By construction the algorithm terminates only at an allocation $\xi$ with $F(\xi) = \emptyset$. Since $\xi$ is attainable and $F(\xi)$ consists of the grants at which (1) fails, $\xi$ is non-wasteful.

**Axiom B (respecting priorities).** Let $\xi = C^f_\Delta(\mathcal{G})$. Two facts from the proof of attainability are used repeatedly: funding never falls along the run, and every $\xi^t$ is attainable, so a grant's first increment is at least its minimum cost.

For the second clause, fix $j \in I$ with $\underline{b}_j \leq \xi_j < \bar{b}_j$, and suppose for contradiction that some $i \in I$ has $j \succ i$, $\rho(j) \supseteq \rho(i)$ and $\xi_i > 0$. Then $i$ is the grant $g$ selected at some step $t$.

- If $\mathscr{C}^t \neq \emptyset$, then $g \in A_c(\xi^{t-1})$ for some $c \in \mathscr{C}^t \cap \rho(g)$. Since $\rho(g) \subseteq \rho(j)$ and $\xi^{t-1}_j \leq \xi_j < \bar{b}_j$, also $j \in A_c(\xi^{t-1})$, and $j \succ g$ contradicts the choice of $g$.
- If $\mathscr{C}^t = \emptyset$, then $g$ is the highest-merit element of $F(\xi^{t-1})$. We show $j \in F(\xi^{t-1})$, which with $j \succ g$ is a contradiction. Step $t$ assigns $g$ a positive amount, so $s(\xi^{t-1}) > 0$. If $\xi^{t-1}_j > 0$, then $\underline{b}_j \leq \xi^{t-1}_j < \bar{b}_j$, and $j \in F(\xi^{t-1})$. If $\xi^{t-1}_j = 0$, then $j$ is first funded at some later step $t'$, where it receives at least $\underline{b}_j$, so $\sum_k \xi^{t-1}_k + \underline{b}_j \leq \sum_k \xi^{t'}_k \leq \mathcal{B}$, that is, $\underline{b}_j \leq s(\xi^{t-1})$, and again $j \in F(\xi^{t-1})$.

For the first clause, fix $j \in I$ with $\xi_j = 0$, and suppose for contradiction that there is a nonempty $H \subseteq I$ with $\xi_h > 0$, $j \succ h$ and $\rho(j) \supseteq \rho(h)$ for every $h \in H$, and

$\sum_{h\in H}\xi_h + s(\xi) \geq \underline{b}_j$. Let $t$ be the first step at which some grant of $H$ is funded, so that the grant $g$ selected at step $t$ lies in $H$ and $\xi_h^{t-1} = 0$ for every $h \in H$.

- If $\mathscr{C}^t \neq \emptyset$, then $g \in A_c(\xi^{t-1})$ for some $c \in \mathscr{C}^t \cap \rho(g)$. Since $c \in \rho(g) \subseteq \rho(j)$ and $\xi_j^{t-1} = 0 < \bar{b}_j$, also $j \in A_c(\xi^{t-1})$, and $j \succ g$ contradicts the choice of $g$.
- If $\mathscr{C}^t = \emptyset$, then $g$ is the highest-merit element of $F(\xi^{t-1})$. All funding of the grants in $H$ is assigned at step $t$ or later, so $\sum_{h\in H}\xi_h \leq \sum_k \xi_k - \sum_k \xi_k^{t-1} = s(\xi^{t-1}) - s(\xi)$. Hence $\underline{b}_j \leq \sum_{h\in H}\xi_h + s(\xi) \leq s(\xi^{t-1})$, and since $\xi_j^{t-1} = 0$, $j \in F(\xi^{t-1})$. Again $j \succ g$ contradicts the choice of $g$.

**Monotonicity in cost.** Let $\xi = C_\Delta^f(\mathcal{G})$, and fix a grant $i^* \in I$. Fix a deviation consisting of an increase in cost for $i^*$. Let $\mathcal{G}'$ be equal to $\mathcal{G}$ except having this deviation, and let $\xi' = C_\Delta^f(\mathcal{G}')$.

**Claim 2.1.** ***(No gain from disimprovement.)*** *If $\xi_{i^*} = 0$, then $\xi'_{i^*} = 0$.*

*Proof.* Write $\underline{b}'_{i^*} \geq \underline{b}_{i^*}$ and $\bar{b}'_{i^*} \geq \bar{b}_{i^*}$ for the cost bounds of $i^*$ in $\mathcal{G}'$; every other component of the two problems coincides, including the merit order. Assume $\xi_{i^*} = 0$. We show by induction that the two runs coincide step by step, which gives $\xi' = \xi$ and in particular $\xi'_{i^*} = 0$. We first record three facts about the run of $C_\Delta^f$ on $\mathcal{G}$.
*(P1) Every non-terminal step assigns a strictly positive amount.* The increment is $\min\{\bar{b}_g - \xi_g^{t-1}, M_t\}$. The first term is positive because $g$ is selected from $\bigcup_{c\in\mathscr{C}^t} A_c(\xi^{t-1})$ or from $F(\xi^{t-1})$, neither of which contains a fully funded grant. If $\mathscr{C}^t \neq \emptyset$, then $M_t \geq d_c(\xi^{t-1}) > 0$ for any $c \in \mathscr{C}^t \cap \rho(g)$, since Axiom C can fail at $c$ only when $d_c(\xi^{t-1}) > 0$. If $\mathscr{C}^t = \emptyset$, then $g \in F(\xi^{t-1})$, so $M_t = s(\xi^{t-1}) > 0$.
*(P2) A grant's first funding is at least its minimum cost.* Suppose $\xi_g^{t-1} = 0$. If $\mathscr{C}^t \neq \emptyset$, then $g$ is the highest-merit element of $\bigcup_{c\in\mathscr{C}^t} A_c(\xi^{t-1})$, so for any $c \in \mathscr{C}^t \cap \rho(g)$ it is the highest-merit element of $A_c(\xi^{t-1})$. Axiom C fails at $c$ and $\xi_g^{t-1} = 0$, so $\underline{b}_g \leq d_c(\xi^{t-1}) \leq M_t$. If $\mathscr{C}^t = \emptyset$, membership of $g$ in $F(\xi^{t-1})$ gives $\underline{b}_g \leq s(\xi^{t-1}) = M_t$. In either case the increment $\min\{\bar{b}_g, M_t\}$ is at least $\underline{b}_g$. Since funding never falls, every grant funded at some step $t$ satisfies $\xi_i^{t'} \geq \underline{b}_i$ at all later steps $t' \geq t$.
*(P3) At every non-terminal step $t$, $g \neq i^*$ and $g \in F(\xi^{t-1})$.* By (P1), $\xi_g > 0 = \xi_{i^*}$, so $g \neq i^*$. If $\xi_g^{t-1} = 0$, then by (P2) and the attainability of $\xi$ (established above in the proof of finite termination and attainability), $\sum_i \xi_i^{t-1} + \underline{b}_g \leq \sum_i \xi_i^t \leq \sum_i \xi_i \leq \mathcal{B}$, that is, $\underline{b}_g \leq s(\xi^{t-1})$. If $\xi_g^{t-1} > 0$, then $\underline{b}_g \leq \xi_g^{t-1} < \bar{b}_g$ by (P2) and the selection rule, and $\sum_i \xi_i^{t-1} < \sum_i \xi_i^t \leq \mathcal{B}$ by (P1), that is, $s(\xi^{t-1}) > 0$. Either way $g \in F(\xi^{t-1})$.

Now fix a step $t$ of $C_\Delta^f(\mathcal{G})$ and assume $\xi'^{t-1} = \xi^{t-1} =: \eta$. Since funding never falls and $\xi_{i^*} = 0$, $\eta_{i^*} = 0$. Write $F(\eta)$ and $F'(\eta)$ for the set $F(\eta)$ in $\mathcal{G}$ and in $\mathcal{G}'$, and $U$ and $U'$ for the sets of categories at which $\eta$ fails Axiom C in $\mathcal{G}$ and $\mathcal{G}'$.

*The sets $F(\eta)$ and $F'(\eta)$.* Membership of any $i \neq i^*$ depends only on $\eta$ and $i$'s own data, so it is the same in both problems. Since $\eta_{i^*} = 0$, $i^* \in F'(\eta)$ means $\underline{b}'_{i^*} \leq s(\eta)$, which implies $\underline{b}_{i^*} \leq s(\eta)$. Hence $F'(\eta) \subseteq F(\eta)$, and the two sets differ at most in $i^*$.
*Axiom C.* Fix a category $c$. The shortfall $d_c(\eta)$ depends only on $\eta$ and on eligibility, so it is the same in both problems; if $d_c(\eta) \leq 0$, Axiom C holds at $c$ in both. Suppose $d_c(\eta) > 0$. The set $A_c(\eta)$ is also the same in both problems: membership of $i \neq i^*$ is unchanged, and if $c \in \rho(i^*)$ then $\eta_{i^*} = 0 < \bar{b}_{i^*} \leq \bar{b}'_{i^*}$, so $i^*$ belongs to $A_c(\eta)$ in both. As merit is unchanged, $A_c(\eta)$ has the same highest-merit element $i^\circ$ in both problems. If $i^\circ \neq i^*$, the test of Axiom C at $c$ involves only $\eta_{i^\circ}$, $\underline{b}_{i^\circ}$ and $d_c(\eta)$, and returns the same verdict in both problems. If $i^\circ = i^*$, the test reads $\underline{b}_{i^*} > d_c(\eta)$ in $\mathcal{G}$ and $\underline{b}'_{i^*} > d_c(\eta)$ in $\mathcal{G}'$, and the first implies the second. Therefore

$$U' \subseteq U, \qquad \text{and} \qquad c \in U \setminus U' \text{ only if the highest-merit element of } A_c(\eta) \text{ is } i^*. \tag{6}$$

*Termination.* If step $t$ is terminal for $\mathcal{G}$, then $F(\eta) = \emptyset$, so $F'(\eta) = \emptyset$ and the run on $\mathcal{G}'$ terminates at the same step with the same vector $\eta$. Otherwise, by (P3), $g \in F(\eta)$ and $g \neq i^*$, so $g \in F'(\eta)$ and step $t$ is non-terminal for $\mathcal{G}'$ as well. It remains to show that the same grant receives the same amount.
*Case $U = \emptyset$.* By (6), $U' = \emptyset$, so $\mathscr{C}^t = \mathscr{C}'^t = \emptyset$ and the runs select the highest-merit elements of $F(\eta)$ and $F'(\eta)$. Since $g$ is the highest-merit element of $F(\eta)$ and belongs to the subset $F'(\eta)$, it is also the highest-merit element of $F'(\eta)$. The amount $M_t = s(\eta)$ is the same in both problems, and $\bar{b}_g$ is unchanged because $g \neq i^*$.
*Case $U \neq \emptyset$.* Then $\mathscr{C}^t \neq \emptyset$ and $g$ is the highest-merit element of $\bigcup_{c \in \mathscr{C}^t} A_c(\eta)$. In particular $\mathscr{C}^t \cap \rho(g) \neq \emptyset$, and for every $c \in \mathscr{C}^t \cap \rho(g)$ the highest-merit element of $A_c(\eta)$ is $g \neq i^*$. (This is the sense in which $g$ outranks $i^*$: if $i^*$ belongs to $\bigcup_{c \in \mathscr{C}^t} A_c(\eta)$, then $g \succ i^*$, and in any case $i^*$ is not the next-best grant eligible for any category in $\mathscr{C}^t \cap \rho(g)$.) We show:

(a) $\rho(g) \cap \mathscr{C}'^t = \rho(g) \cap \mathscr{C}^t$, and

(b) $\mathscr{C}'^t \subseteq \mathscr{C}^t$.

By (6), every $c \in \mathscr{C}^t \cap \rho(g)$ lies in $U'$. Groups of $\Delta$ with smaller index than $D^t$ contain no category of $U$, hence none of $U'$, while $D^t$ contains a category of $\mathscr{C}^t \cap \rho(g) \subseteq U'$. So the active group in $\mathcal{G}'$ is also $D^t$, and $\mathscr{C}'^t = U' \cap D^t \subseteq U \cap D^t = \mathscr{C}^t$, which is (b). Since $\mathscr{C}^t \cap \rho(g) \subseteq U' \cap D^t = \mathscr{C}'^t$, (b) gives (a).

The sets $A_c(\eta)$ coincide in the two problems, so by (b), $\bigcup_{c \in \mathscr{C}'^t} A_c(\eta) \subseteq \bigcup_{c \in \mathscr{C}^t} A_c(\eta)$, and by (a) the smaller union still contains $g$. Being the highest-merit element of the larger union, $g$ is the highest-merit element of the smaller one, so the run on $\mathcal{G}'$ also selects $g$ at step $t$. By (a), and since $g \neq i^*$,

$$\xi^t_g = \eta_g + \min\left\{\bar{b}_g - \eta_g, \max_{c \in \mathscr{C}^t \cap \rho(g)} d_c(\eta)\right\} = \eta_g + \min\left\{\bar{b}_g - \eta_g, \max_{c \in \mathscr{C}'^t \cap \rho(g)} d_c(\eta)\right\} = \xi'^t_g.$$

In both cases the same grant $g$ receives the same amount and no other grant's funding changes, so $\xi'^t = \xi^t$. By induction the two runs coincide and terminate together, so $\xi' = \xi$, and $\xi'_{i^*} = \xi_{i^*} = 0$. □

**Claim 2.2.** ***(No harm from improvement.)*** *If $\xi'_{i^*} > 0$, then $\xi_{i^*} > 0$.*

*Proof.* This follows directly from Claim 2.1. Since $\xi'_{i^*} = 0$ whenever $\xi_{i^*} = 0$, the contrapositive holds: $\neg(\xi_{i^*} = 0)$ whenever $\neg(\xi'_{i^*} = 0)$. But this is equivalent to Claim 2.2, $\xi_{i^*} > 0$ whenever $\xi'_{i^*} > 0$. □

**Failure of monotonicity in merit and categories.** See Example 6.

**Independence of non-contending grants.** Consider a floor problem $\mathcal{G}$ with grants $I$, and a floor problem $\mathcal{G}^{+j}$ with grants $I \cup \{j\}$, which is the same but adds a new grant $j$ which is non-contending at $\xi$. In the floor phase of $C^f_\Delta$, since the grant $j$ is non-marginal, it does not affect whether feasible floors are satisfied for any category. Moreover, in each of its categories there is always a grant with higher merit that is not fully funded, so the stopping rule for consideration of each category dictates that $j$ is never funded in this phase. We argue by induction on the steps of the algorithm that the runs on $\mathcal{G}$ and $\mathcal{G}^{+j}$ produce the same sequence $\xi^0, \xi^1, \ldots$, with $\xi^t_j = 0$ throughout. Suppose they agree through step $t-1$. Since funding only increases along the algorithm, the second condition of non-marginality at the final allocation $\xi$ holds a fortiori at $\xi^{t-1}$: in every $c \in \rho(j)$ some $i \succ j$ has $\xi^{t-1}_i < \bar{b}_i$. Hence $j$ is never the highest-merit grant in $A_c(\xi^{t-1})$, so adding $j$ alters neither which categories satisfy Axiom C, and thus $\mathscr{C}^t$, nor the grant $g$ chosen when $\mathscr{C}^t \neq \emptyset$. If $\mathscr{C}^t = \emptyset$ and $F(\xi^{t-1})$ is nonempty in $\mathcal{G}$, the grant $g$ selected there receives a positive amount, as at every non-terminal step (see (P1) above); so $\xi_g > 0$ and, $j$ being non-contending, $g \succ j$. The set $F(\xi^{t-1})$ in $\mathcal{G}^{+j}$ differs from that in $\mathcal{G}$ at most by $j$, so the same $g$ is selected. Finally, when the run on $\mathcal{G}$ terminates at $\xi$, $F(\xi)$ is empty, and by the first condition of non-marginality $\underline{b}_j > s(\xi)$, so $j \notin F(\xi)$ in $\mathcal{G}^{+j}$ either. Both runs therefore terminate at the same step, and $C^f_\Delta(\mathcal{G}^{+j})$ coincides with $C^f_\Delta(\mathcal{G})$ on $I$ and assigns $j$ nothing. □

*Proof of Proposition 6.* Fix an ordered partition $\Delta$, a floor problem $\mathcal{G}$, and a grant $j \in I$ that is non-marginal at $\xi := C^f_\Delta(\mathcal{G})$. Let $\theta_j \succsim_t \theta'_j$ and let $\mathcal{G}'$ be $\mathcal{G}$ with $j$'s report replaced by $\theta'_j$.

Non-marginality is defined only at grants with $\xi_j = 0$, and since funding is non-decreasing along the run, $\xi^t_j = 0$ at every step. By the second condition of non-marginality, for each $c \in \rho(j)$ fix a witness for $c$: a grant $w \in I_c$ with $w \succ j$ and $\xi_w < \bar{b}_w$. Again by monotonicity of the run, $\xi^t_w < \bar{b}_w$ at every step, and since $w \neq j$, the report of $w$ is unchanged in $\mathcal{G}'$.

We show by induction that the two runs coincide. They agree at $\xi^0 = \xi'^0 = 0$. Suppose they agree through step $t-1$, and write $\eta$ for the common funding vector, so $\eta_j = 0$.

**(a) The satisfied categories are the same.** Fix $c$. If $c \notin \rho(j)$, nothing bearing on $c$ differs between the problems, so assume $c \in \rho(j)$ and let $w$ be its witness. Check Axiom C at $c$. The shortfall is the same under both problems: the only term of $\sum_{i \in I_c} \eta_i$ that could differ is $\eta_j = 0$, whether or not $j$ has dropped $c$, so $d_c(\eta)$ is unchanged and in particular $d_c(\eta) > 0$ has the same truth value. Suppose $d_c(\eta) > 0$. Under $\mathcal{G}$ the set $A_c(\eta)$ contains $w$, since $w \in I_c$ and $\eta_w < \bar{b}_w$, and $w \succ j$; so the highest-merit element of $A_c(\eta)$ is not $j$. Passing to $\mathcal{G}'$ changes only $j$'s data: it may delete $j$ from $A_c(\eta)$ or lower its merit, but leaves $w$ above $j$, so the highest-merit element of $A_c(\eta)$ and its minimum cost are unchanged. Hence Axiom C holds at $c$ under $\mathcal{G}$ if and only if it holds under $\mathcal{G}'$, and $\mathscr{C}^t$ and $D^t$ coincide.

**(b) The same grant is funded, by the same amount.** Suppose $\mathscr{C}^t \neq \emptyset$. The algorithm selects the highest-merit element of $\bigcup_{c \in \mathscr{C}^t} A_c(\eta)$. If $j$ belongs to this set under $\mathcal{G}$, then $c \in \mathscr{C}^t \cap \rho(j)$ for some $c$, and the witness $w$ for $c$ lies in the same set with $w \succ j$; so $g \neq j$. Passing to $\mathcal{G}'$ can only delete $j$ from the set or lower its merit, leaving every other element untouched, so $g$ is unchanged. The amount $M_t$ depends only on the floors, on the totals $\sum_{i \in I_c} \eta_i$, which are unchanged by (a), and on $\rho(g)$ with $g \neq j$, so the increment $\min\{\bar{b}_g - \eta_g, M_t\}$ is unchanged.

Suppose instead $\mathscr{C}^t = \emptyset$. The algorithm selects the highest-merit element of $F(\eta)$. Were that $j$, grant $j$ would receive funding at step $t$, contradicting $\xi_j = 0$; so $g \neq j$. Under $\mathcal{G}'$, $F(\eta)$ can differ only in whether it contains $j$, since $\underline{b}'_j \geq \underline{b}_j$ and no other grant's parameters change, and $j$ was not the maximum in any case. So $g$ is unchanged, as is $M_t = \mathcal{B} - \sum_i \eta_i$.

**(c) The runs stop together.** $C^f_\Delta(\mathcal{G})$ halts at the first step at which $F(\eta)$ is empty. At that step, membership under $\mathcal{G}'$ can only be more restrictive: for $j$, exclusion under $\mathcal{G}$ means $\underline{b}_j > s(\eta)$, and $\underline{b}'_j \geq \underline{b}_j$; every other grant is unchanged. So $C^f_\Delta(\mathcal{G}')$ halts no later than $C^f_\Delta(\mathcal{G})$.

It remains to rule out an earlier halt under $\mathcal{G}'$. The two sets $F(\eta)$ differ only in whether they contain $j$, so at a step where $\mathcal{G}$ does not halt, $\mathcal{G}'$ could halt only if $F(\eta) = \{j\}$ in $\mathcal{G}$. Suppose it were. Then $s(\eta) \geq \underline{b}_j > 0$, so any $i \neq j$ with $\underline{b}_i \leq \eta_i < \bar{b}_i$ would itself lie in $F(\eta)$; hence every $i \neq j$ is either fully funded or has $\eta_i = 0$ and $\underline{b}_i > s(\eta)$. If $\mathscr{C}^t = \emptyset$, the algorithm selects the highest-merit element of $F(\eta) = \{j\}$, giving $g = j$ and contradicting (b). If $\mathscr{C}^t \neq \emptyset$, then $g \neq j$ by (b) and $\eta_g < \bar{b}_g$, so $\eta_g = 0$ and $\underline{b}_g > s(\eta)$. Grant $g$ receives funding at step $t$, so $\xi_g > 0$ and therefore $\xi_g \geq \underline{b}_g$; since funding is nondecreasing along the run, $\sum_i \xi_i \geq \sum_i \eta_i + \underline{b}_g > \mathcal{B}$, contradicting the attainability of $\xi$ established in the proof of Theorem 2. So $F(\eta) \neq \{j\}$ at every non-terminal step, and the two runs halt together with the same allocation. □

*Proof of Proposition 7.* **Step bound.** Every non-terminal step increases the funding of exactly one grant, $g$, by $\min\{\bar{b}_g - \xi_g^{t-1}, M_t\}$, and this quantity is strictly positive: the selection rule gives $\xi_g^{t-1} < \bar{b}_g$, and $M_t > 0$ because either $\mathscr{C}^t \cap \rho(g) \neq \emptyset$, in which case any such category fails Axiom C and hence has $d_c(\xi^{t-1}) > 0$, or $\mathscr{C}^t = \emptyset$ and $g \in F(\xi^{t-1})$, so $M_t = s(\xi^{t-1}) > 0$. Whichever term attains the minimum then determines which of three events occurs. If $\bar{b}_g - \xi_g^{t-1} \leq M_t$, grant $g$ becomes fully funded; since funding never falls and fully funded grants are excluded from later selection, this occurs at most $|I|$ times. If $M_t$ attains the minimum and $\mathscr{C}^t \neq \emptyset$, let $e$ attain the maximum defining $M_t$; then $g \in I_e$ and $d_e(\xi^t) = 0$, so $e$ becomes satisfied, and by Lemma 2 it remains so; this occurs at most $|\mathscr{C}|$ times, and it also implies that the index of the active group $D^t$ is nondecreasing. If $M_t$ attains the minimum and $\mathscr{C}^t = \emptyset$, the budget is exhausted and the algorithm terminates at the next step; this occurs at most once, and if $g$ is not fully funded at that step it never is, so this event and full funding together occur at most $|I|$ times. Hence there are at most $|I| + |\mathscr{C}|$ non-terminal steps, and $T \leq |I| + |\mathscr{C}| + 1$.

**Per-step cost.** The implementation stores $\xi$ and the running totals $S_c = \sum_{i \in I_c} \xi_i$. Testing Axiom C at a category $c$ needs $S_c$ and the highest-merit element of $A_c(\xi)$, found by scanning the merit-ordered list $I_c$; so constructing $D^t$ and $\mathscr{C}^t$ costs $O(\sum_c |I_c|) \subseteq O(|I| \cdot |\mathscr{C}|)$, and selecting $g$ costs no more. Given the totals retained from that sweep, computing $M_t$ and updating $\xi_g$ and $(S_c)_{c \in \rho(g)}$ costs $O(|\mathscr{C}|)$. Multiplying $O(|I| \cdot |\mathscr{C}|)$ per step by the step bound gives the result. □

## A.3 Merit-Maximal Choice Rules

*Proof of Theorem 3.* The merit-maximal rule $C^{\mathrm{mm}}_{\mathcal{M}}$ is uniquely determined, since $\Xi(\mathcal{G})$ is a finite non-empty totally ordered set and therefore has a single maximum. It suffices, then, to show that $C^{\mathrm{mm}}_{\mathcal{M}}$ itself fails each of the listed properties: any $C \in \mathcal{M}$ possessing one of them differs from $C^{\mathrm{mm}}_{\mathcal{M}}$ and so is not merit-maximal on $\mathcal{M}$. Independence of non-contending grants fails by Example 7, and monotonicity in merit and in categories by Example 6. It remains to treat monotonicity in cost, and monotonicity in types for non-marginal grants. To do this, we consider Example 5 and construct a monotonic disimprovement for a grant which is non-marginal.

The first step is finding the allocation $C^{\mathrm{mm}}_{\mathcal{M}}(\mathcal{G}) = (4, 12, 0)$. We first check $(4, 12, 0) \in \Xi(\mathcal{G})$. It exhausts the budget, so it is non-wasteful (Axiom A). Its category total is $12 \geq q_\alpha$, so $d_\alpha(\xi) \leq 0$ and Axiom C is satisfied. For Axiom B, the grants funded short of their maximum costs are grants 1 and 2: grant 1 has no lower-priority grant, since $\rho(1) = \emptyset \not\supseteq \{\alpha\}$, and the only lower-priority grant of grant 2 is grant 3, which is unfunded, so the second clause holds. The only unfunded grant, grant 3, has no funded grant below it, so the first clause holds as well.

No allocation with $\xi_1 > 4$ lies in $\Xi(\mathcal{G})$. Suppose $\xi_1 > 4$, so that $d_\alpha(\xi) = 12 - \xi_2 -$

$\xi_3 > 0$ and Axiom C must therefore bind. Since $\xi_2 \leq \mathcal{B} - \xi_1 < 12 < 13 = \bar{b}_2$, grant 2 is not fully funded, so grant 2 is the highest-merit element of $A_\alpha(\xi)$, and Axiom C requires $\xi_2 = 0$ together with $\underline{b}_2 > d_\alpha(\xi) = 12 - \xi_3$, i.e. $\xi_3 > 5$. But $\bar{b}_3 = 4$, so this is impossible. (Concretely, the only non-wasteful allocation with $\xi_1 = 10$ is $(10, 0, 4)$: at $(10, 0, 3)$ or below, grant 3 can still absorb unassigned funds, while at $(10, 0, 4)$ the two unassigned units cannot be placed anywhere. And $(10, 0, 4)$ fails Axiom C, because $\underline{b}_2 = 7 \leq 8 = d_\alpha(\xi)$.) Among allocations with $\xi_1 = 4$, the budget permits at most $\xi_2 = 12$, and $(4, 12, 0)$ therefore dominates every other element of $\Xi(\mathcal{G})$.

We will show that grant 3 is non-marginal at $C^{\text{mm}}_{\mathcal{M}}(\mathcal{G})$. Write $\xi = (4, 12, 0)$. Condition 1 holds: $\underline{b}_3 = 2 > 0 = s(\xi)$. Condition 2 holds: $\rho(3) = \{\alpha\}$, and grant 2 is eligible for $\alpha$, has higher merit than grant 3, and has $\xi_2 = 12 < 13 = \bar{b}_2$. (Grant 3 is moreover non-contending at $\xi$, since the funded grants 1 and 2 both outrank it; this is not needed here, but it is what the independence property would additionally require.) So grant 3 sits at neither margin: the budget is exhausted and it cannot absorb the residual, and $\alpha$, the only category it is eligible for, has a higher-merit eligible grant that is not yet fully funded.

It remains to construct a monotonic deviation. Let $\mathcal{G}'$ be identical to $\mathcal{G}$ except that grant 3 reports $[\underline{b}'_3, \bar{b}'_3] = [2, 6]$. Since $\underline{b}'_3 = \underline{b}_3$, $\bar{b}'_3 > \bar{b}_3$, the claimed categories are unchanged, and $m'_3 \leq m_3$, we have $\theta_3 \succsim_t \theta'_3$: the deviation is a weak disimprovement. If $m'_3 = m_3$, the disimprovement is in cost alone.

Then $C^{\text{mm}}_{\mathcal{M}}(\mathcal{G}') = (10, 0, 6)$. This allocation exhausts the budget, so Axiom A holds. Its category total is $6 < q_\alpha$, so $d_\alpha(\xi) = 6$; the highest-merit element of $A_\alpha(\xi)$ is grant 2, with $\xi_2 = 0$ and $\underline{b}_2 = 7 > 6$, so Axiom C is satisfied. For Axiom B, grants 1 and 3 are fully funded, so the second clause of Axiom B is vacuous, and the only unfunded grant is grant 2, whose sole lower-priority funded grant contributes $\xi_3 = 6$; with no unspent budget, $6 < 7 = \underline{b}_2$. Since $\bar{b}_1 = 10$, no allocation gives grant 1 more, and $(10, 0, 6)$ is the only non-wasteful allocation with $\xi_1 = 10$; hence it is the maximum of $\Xi(\mathcal{G}')$.

Therefore $C^{\text{mm}}_{\mathcal{M}}(\mathcal{G})_3 = 0$ while $C^{\text{mm}}_{\mathcal{M}}(\mathcal{G}')_3 = 6 > 0$, with grant 3 non-marginal at $C^{\text{mm}}_{\mathcal{M}}(\mathcal{G})$. Monotonicity in types therefore fails, and under the reading in which $m'_3 = m_3$, so does monotonicity in cost. □

## A.4 Strong Feasible Floors

Recall Axiom C* from Section 4.4: whenever category $c$ falls short of its floor, every grant eligible for $c$ that is not fully funded, that is, every grant in $A_c(\xi)$, is unfunded and has a minimum cost exceeding the shortfall $d_c(\xi)$, so that funding any of them would carry $c$ past its floor. Since the binding case is the cheapest eligible grant that is not fully funded, C* demands more funding before a floor counts as met than Axiom C does. We first prove Proposition 1, stated in Appendix 4.5, and then turn

to whether Axioms A, B and C* can always be satisfied together.

*Proof of Proposition 1.* Suppose there are four grants $1 \succ 2 \succ 3 \succ 4$, given by:

$$\begin{array}{lll} 1: & [\underline{b}_1, \bar{b}_1] = [2,2], & \rho(1) = \emptyset \\ 2: & [\underline{b}_2, \bar{b}_2] = [1,1], & \rho(2) = \{\alpha\} \\ 3: & [\underline{b}_3, \bar{b}_3] = [4,4], & \rho(3) = \{\alpha\} \\ 4: & [\underline{b}_4, \bar{b}_4] = [2,2], & \rho(4) = \{\alpha\}, \end{array}$$

where $\mathscr{C} = \{\alpha\}$, $\mathcal{B} = 4$, and $q_\alpha = 3$. Consider the grant-funding problem which omits the lowest merit grant 4. The unique allocation $\xi$ satisfying non-wastefulness, respecting priorities, and satisfying strong feasible floors (Axioms A, B and C*) is $(2,1,0)$. It is worth seeing why $(2,1,0)$ satisfies both floor axioms before grant 4 arrives. The category is short by $d_\alpha(\xi) = 3 - 1 = 2$, and the only eligible grant that is not fully funded is grant 3, at a cost of 4. Since $4 > 2$, funding grant 3 would overshoot the floor, so the category counts as satisfied under Axiom C and, grant 3 being the only candidate, under Axiom C* as well. Adding grant 4 changes nothing for Axiom C, whose threshold is still set by the highest-merit candidate grant 3; but grant 4 costs 2, so under Axiom C* the category is no longer satisfied, and closing the deficit forces grant 1 out. In the problem which includes grant 4, the unique solution satisfying Axioms A, B, and C* is $(0,1,0,2)$. Grant 4 is non-marginal at $(2,1,0)$—its minimum cost 2 exceeds the residual 1, and $\alpha$, the only category it is eligible for, has a higher-merit eligible grant, grant 3, that is unfunded and so not fully funded—and it is non-contending there, since the funded grants 1 and 2 both outrank it. Therefore, we have exhibited a non-contending grant which affects the allocation. □

Compare the allocation to the floors-first choice rule (which satisfies Axioms A, B and C). We have $C^f_\Delta(\mathcal{G}) = (2,1,0)$ when grant 4 is omitted, and $C^f_\Delta(\mathcal{G}^{+4}) = (2,1,0,0)$ when grant 4 is included; the allocation is unaffected since $C^f_\Delta$ is independent of non-contending grants. The violation arises because strong feasible floors force the allocation to "squeeze in" additional funding for a category whenever doing so closes its floor deficit, even at the expense of higher-merit grants. The contrast between Axioms C* and C parallels that drawn by Delacrétaz et al. (2023) between envy-freeness, which rules out every priority violation, and interference-freeness, which rules out only those in which a higher-priority family could claim capacity used by the envied family (see Section 2). Like envy-freeness, Axiom C* tests the allocation against every eligible grant; like interference-freeness, Axiom C tests it only at the relevant margin.

### Joint satisfiability

Whether Axioms A, B and C* can be met simultaneously in every floor problem is a question we leave open. For Axioms A–C, existence follows from the construction of $C^f_\Delta$ (Theorem 2), which advances an unsatisfied category by funding its highest-merit eligible grant that is not fully funded, a step that never creates a priority violation. This argument does not extend to Axiom C*, under which a floor is met only once $d_c(\xi)$ falls below the minimum cost of every grant in $A_c(\xi)$; the threshold is thus set by the cheapest such grant, which may sit anywhere in the merit order. Meeting it may require funding a grant outranked under $\succ_\rho$ by an unfunded eligible grant, which Axiom B forbids once the funds committed, together with the unspent budget, reach that grant's minimum cost, or whenever an outranking grant is funded short of its maximum cost. The proof of Proposition 1 illustrates both the tension and its limits. Axiom B blocks meeting the threshold through grant 3, since funding grant 3 would leave grant 2 unfunded with $\xi_3 \geq \underline{b}_2$; the funds can instead be drawn from grant 1, which is incomparable under $\succ_\rho$ to every grant eligible for $\alpha$ because $\rho(1) = \emptyset$. A counterexample to joint satisfiability would have to close this route as well.

We have not been able to derive an impossibility from this tension. We instead report the results of an exhaustive computational search for counterexamples.[5] Fixing a number of grants, a number of categories and a budget, the search enumerates every integer floor problem—cost bounds $1 \leq \underline{b}_i \leq \bar{b}_i \leq \mathcal{B}$, eligibility sets ranging over all subsets of $\mathscr{C}$, and floors $q_c \geq 1$ with $\sum_c q_c \leq \mathcal{B}$—and for each enumerates every integer allocation, testing Axioms A, B and C* directly against their statements in Sections 4.3 and 4.4; for Axiom C* this means checking, at every category carrying a positive shortfall, that every grant in $A_c(\xi)$ is unfunded and has a minimum cost exceeding $d_c(\xi)$. A problem is flagged if and only if no integer allocation satisfies all three axioms. Grants are indexed by merit position, $1 \succ \cdots \succ n$, and cost bounds and eligibility are assigned to each position independently. Since the axioms refer to grants only through their data and the merit order, every problem of these dimensions with positive floors is a relabeling of exactly one problem in this class, so fixing the order entails no loss of generality, and the counts below are of problems up to relabeling. Zero floors are excluded: a category with no floor never binds Axiom C*, but its eligibility still enters the comparisons of Axiom B, so problems with such a category lie outside the class searched.

[5]The search code, together with the configurations reported below, is part of the same deposit (Jourden, 2026) and is also included in the supplementary material; a development copy is at `https://github.com/joejourden/priority-design-grant-funding`. The code was written with the assistance of Claude (Anthropic), a generative AI tool, and reviewed by the author; its README explains how each axiom test follows the statement of the axiom and how the optimized enumeration is checked against a direct search.

No problem was flagged. Across all fourteen configurations $(|I|, |\mathscr{C}|, \mathcal{B})$—$(2,1,10)$, $(2,2,9)$, $(2,3,8)$, $(3,1,8)$, $(3,2,8)$, $(3,3,7)$, $(4,1,7)$, $(4,2,6)$, $(4,3,5)$, $(5,1,5)$, $(5,2,4)$, $(5,3,3)$, $(6,1,4)$ and $(6,2,3)$, approximately $5.2 \times 10^9$ problems up to relabeling of grants—every problem admits at least one allocation satisfying all three axioms, as does each of twelve million problems drawn at random with up to nine grants, four categories and budgets up to forty. In these draws the number of grants, the budget and the number of categories are uniform on $\{2, \ldots, 9\}$, $\{2, \ldots, 40\}$ and $\{1, \ldots, \min(4, \mathcal{B})\}$; cost bounds and eligibility are drawn independently for each merit position; and the floors are uniform over $\{q : q_c \geq 1,\ \sum_c q_c \leq \mathcal{B}\}$. At these budgets, exhaustive enumeration of allocations is computationally prohibitive, so the allocations of each drawn problem are instead searched depth-first over the same space: grants are assigned integer funding levels one at a time in merit order, a branch is abandoned only once its partial allocation exceeds the budget, each completed allocation is tested against Axioms A, B and C$^*$ exactly as in the exhaustive search, and the search stops at the first allocation that passes. The only pruning discards vectors that are not allocations, so a drawn problem is flagged if and only if it has no integer allocation satisfying all three axioms. Each drawn problem is therefore resolved exactly, as in the exhaustive search; randomization enters only through the selection of problems, not through the search over allocations. We note two further properties of the search. First, the axioms are invariant to a common rescaling of the budget, the cost bounds and the floors, so each configuration covers every problem with its numbers of grants and categories and with positive floors whose data are commensurable and can be rescaled to integers with exactly the budget searched; smaller budgets are not enumerated. Second, since an integer allocation satisfying the axioms establishes existence, restricting the search to integer allocations could only produce spurious flags, and none arose.

The configurations searched are small relative to the problems a funding body confronts, but they include the scale at which the other defects of strong feasible floors already appear: the failure of independence in Proposition 1 requires only four grants and a single category, whereas no instance of non-existence occurs anywhere in the range searched. Our case against Axiom C$^*$ therefore rests on its failure of independence rather than on any claim that the axioms are jointly unsatisfiable.

# B NIH Review Process

As stated by the National Institutes of Health (NIH, 2021), after grant applications have been matched to appropriate reviewers, the funding decisions are made in two steps: (1) peer review, and (2) advisory council or board review that we summarize

as follows:

(1) Peer review:

   (i) Initial peer review meetings are administered by either the Center for Scientific Review (CSR) or one of the NIH ICs (Institutes/Centers) with the funding authority as specified in the funding opportunity announcement (FOA).

   (ii) Reviewers provide an overall impact score to "reflect their assessment of the likelihood for the project to exert a sustained, powerful influence on the research field(s) involved, in consideration of the following review criteria, and additional review criteria (as applicable for the project proposed): significance, investigator(s), innovation, approach, environment" (as well as additional unscored criteria). This score ranges from 1 (exceptional) to 9 (poor).

   (iii) The final overall impact score for each discussed application is determined by calculating the mean score from all the eligible members' final impact scores, and multiplying the average by 10.

(2) Advisory council or board

   (i) Members are chosen by the respective IC and are approved by the Department of Health and Human Services. For certain committees, members are appointed by the President of the United States.

   (ii) NIH program staff members examine applications and consider the overall impact scores given during the peer review process, percentile rankings (if applicable) and the summary statements in light of the Institute/Center's priorities.

   (iii) The Advisory Council/Board also considers the Institute/Center's goals and needs and advises the Institute/Center director concerning funding decisions.

   (iv) The Institute/Center director makes final funding decisions based on staff and Advisory Council/Board advice.

In short, in the first step, peer review assigns an overall impact score to each discussed application, and in the second step, the advisory council or board determines funding advice based on impact scores with consideration of institute objectives.

These scores are coarser than the strict merit order of our model. Because the overall impact score is a scaled mean of reviewers' integer scores, two discussed applications may receive the same score, and applications that are not discussed receive

no overall impact score at all. The model's merit order $\succ$ is therefore not read off directly from this process; it is a strict ranking that extends the scores to all applications, discussed or not, and it is maintained as an assumption (see the discussion following Theorem 1).

Per the *NIH-Wide Strategic Plan for Fiscal Years 2016–2020*, for these years the NIH had the following objectives for HIV/AIDS research, which we quote directly (NIH, 2015):

> **HIV/AIDS Research Priorities**
>
> - Reduce incidence, including vaccines
> - Safer, easier-to-use therapies
> - Work towards a cure
> - HIV-associated comorbidities, co-infections
> - Cross-cutting areas: basic research, health disparities, training